



Top 37 CAT Quant Based LR Questions With Video Solutions

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Questions

Instructions

There are nine boxes arranged in a 3×3 array as shown in Tables 1 and 2. Each box contains three sacks. Each sack has a certain number of coins, between 1 and 9, both inclusive.

The average number of coins per sack in the boxes are all distinct integers. The total number of coins in each row is the same. The total number of coins in each column is also the same.

	1st column	2nd column	3rd column
1st Row		9	6
2nd Row	2		
3rd Row	8		

Table 1

	1st column	2nd column	3rd column
1st Row	1**	2*	2*
2nd Row	1**	0*	3*
3rd Row	3*	2**	0**

Table 2

Table 1 gives information regarding the median of the numbers of coins in the three sacks in a box for some of the boxes. In Table 2 each box has a number which represents the number of sacks in that box having more than 5 coins. That number is followed by a * if the sacks in that box satisfy exactly one among the following three conditions, and it is followed by ** if two or more of these conditions are satisfied.

- The minimum among the numbers of coins in the three sacks in the box is 1.
- The median of the numbers of coins in the three sacks is 1.
- The maximum among the numbers of coins in the three sacks in the box is 9.

Question 1

How many boxes have at least one sack containing 9 coins?

- 3
- 4
- 5
- 8

The average number of coins per sack in the boxes are all distinct integers. The total number of coins in each row is the same. The total number of coins in each column is also the same.

	1st column	2nd column	3rd column
1st Row		9	6
2nd Row	2		
3rd Row	8		

Table 1

	1st column	2nd column	3rd column
1st Row	1**	2*	2*
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Table 2

The information regarding the median of the numbers of coins in the three sacks in a box for some of the boxes is given in Table 1. In Table 2 each box has a number which represents the number of sacks in that box having more than 5 coins. That number is followed by a * if the sacks in that box satisfy exactly one among the following three conditions, and it is followed by ** if two or more of these conditions are satisfied.

- The minimum among the numbers of coins in the three sacks in the box is 1.
- The median of the numbers of coins in the three sacks is 1.
- The maximum among the numbers of coins in the three sacks in the box is 9.

VIDEO SOLUTION

Question 2

For how many boxes are the average and median of the numbers of coins contained in the three sacks in that box the same?

The average number of coins per sack in the boxes are all distinct integers. The total number of coins in each row is the same. The total number of coins in each column is also the same.

	1st column	2nd column	3rd column		1st column	2nd column	3rd column
1st Row		9	6	1st Row	1**	2*	2*
2nd Row				2nd Row	1**	0*	3*
3rd Row	8			3rd Row	3*	2**	0**

Table 1 Table 2

Information regarding the median of the numbers of coins in the three sacks is given in Table 2. In Table 2 each box has a number which is the number of sacks in that box having more than 5 coins. That number is 1** if the sacks in that box satisfy exactly one among the following three conditions, and it is followed by * if two or more of these conditions are satisfied.

i) The minimum among the numbers of coins in the three sacks in the box is 1.
 ii) The median of the numbers of coins in the three sacks is 1.
 iii) The maximum among the numbers of coins in the three sacks in the box is 9.

VIDEO SOLUTION

Question 3

How many sacks have exactly one coin?

The average number of coins per sack in the boxes are all distinct integers. The total number of coins in each row is the same. The total number of coins in each column is also the same.

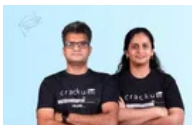
	1st column	2nd column	3rd column		1st column	2nd column	3rd column
1st Row		9	6	1st Row	1**	2*	2*
2nd Row				2nd Row	1**	0*	3*
3rd Row	8			3rd Row	3*	2**	0**

Table 1 Table 2

Information regarding the median of the numbers of coins in the three sacks is given in Table 2. In Table 2 each box has a number which is the number of sacks in that box having more than 5 coins. That number is 1** if the sacks in that box satisfy exactly one among the following three conditions, and it is followed by * if two or more of these conditions are satisfied.

i) The minimum among the numbers of coins in the three sacks in the box is 1.
 ii) The median of the numbers of coins in the three sacks is 1.
 iii) The maximum among the numbers of coins in the three sacks in the box is 9.

VIDEO SOLUTION



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Question 4

What is the total number of coins in all the boxes in the 3rd row?

- A 36
- B 30
- C 15
- D 45

The average number of coins per sack in the boxes are all distinct integers. The total number of coins in each row is the same. The total number of coins in each column is also the same.

	1st column	2nd column	3rd column		1st column	2nd column	3rd column
1st Row		9	6	1st Row	1**	2*	2*
2nd Row				2nd Row	1**	0*	3*
3rd Row	8			3rd Row	3*	2**	0**

Table 1

Table 2

i) The minimum among the numbers of coins in the three sacks in the box is 1.
 ii) The median of the numbers of coins in the three sacks is 1.
 iii) The maximum among the numbers of coins in the three sacks in the box is 9.

VIDEO SOLUTION

Question 5

In how many boxes do all three sacks contain different numbers of coins?

The average number of coins per sack in the boxes are all distinct integers. The total number of coins in each row is the same. The total number of coins in each column is also the same.

	1st column	2nd column	3rd column		1st column	2nd column	3rd column
1st Row		9	6	1st Row	1**	2*	2*
2nd Row				2nd Row	1**	0*	3*
3rd Row	8			3rd Row	3*	2**	0**

Table 1

Table 2

i) The minimum among the numbers of coins in the three sacks in the box is 1.
 ii) The median of the numbers of coins in the three sacks is 1.
 iii) The maximum among the numbers of coins in the three sacks in the box is 9.

VIDEO SOLUTION

Instructions

A train travels from Station A to Station E, passing through stations B, C, and D, in that order. The train has a seating capacity of 200. A ticket may be booked from any station to any other station ahead on the route, but not to any earlier station.

A ticket from one station to another reserves one seat on every intermediate segment of the route. For example, a ticket from B to E reserves a seat in the intermediate segments B - C, C - D, and D - E.

The occupancy factor for a segment is the total number of seats reserved in the segment as a percentage of the seating capacity. The total number of seats reserved for any segment cannot exceed 200.

The following information is known.

1. Segment C - D had an occupancy factor of 95%. Only segment B - C had a higher occupancy factor.
2. Exactly 40 tickets were booked from B to C and 30 tickets were booked from B to E.
3. Among the seats reserved on segment D - E, exactly four-sevenths were from stations before C.
4. The number of tickets booked from A to C was equal to that booked from A to E, and it was higher than that from B to E.
5. No tickets were booked from A to B, from B to D and from D to E.
6. The number of tickets booked for any segment was a multiple of 10.

Question 6

How many tickets were booked from Station A to Station E?

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Question 7

What was the occupancy factor for segment D - E?

- A 35%
- B 70%
- C 77%
- D 84%

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Question 8

How many tickets were booked from Station C?

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Question 9

What is the difference between the number of tickets booked to Station C and the number of tickets booked to Station D?

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Question 10

How many tickets were booked to travel in exactly one segment?

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Instructions

DIRECTIONS for the following questions: These questions are based on the situation given below: A young girl Roopa leaves home with x flowers, goes to the bank of a nearby river. On the bank of the river, there are four places of worship, standing in a row. She dips all the x flowers into the river. The number of flowers doubles. Then she enters the first place of worship, offers y flowers to the deity. She dips the remaining flowers into the river, and again the number of flowers doubles. She goes to the second place of worship, offers y flowers to the deity. She dips the remaining flowers into the river, and again the number of flowers doubles. She goes to the third place of worship, offers y flowers to the deity. She dips the remaining flowers into the river, and again the number of flowers doubles. She goes to the fourth place of worship, offers y flowers to the deity. Now she is left with no flowers in hand.

Question 11

The minimum number of flowers with which Roopa leaves home is:

- A 16
- B 15
- C 0
- D Cannot be determined

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DIRECTIONS for the following questions: These questions are based on the situation given below: A young girl Roopa leaves home with x flowers, goes to the bank of a nearby river. On the bank of the river, there are four places of worship, standing in a row. She dips all the x flowers into the river. The number of flowers doubles. Then she enters the first place of worship, offers y flowers to the deity. She dips the remaining flowers into the river, and again the number of flowers doubles. She goes to the second place of worship, offers y flowers to the deity. She dips the remaining flowers into the river, and again the number of flowers doubles. She goes to the third place of worship, offers y flowers to the deity. She dips the remaining flowers into the river, and again the number of flowers doubles. She goes to the fourth place of worship, offers y flowers to the deity. Now she is left with no flowers in hand.

If Roopa leaves home with 30 flowers, the number of flowers she offers to each deity is:

A) 30
B) 32
C) 31
D) 33

Handwritten notes:

I. $2x - y$
II. $4x - 3y$
III. $8x - 6y$
IV. $16x - 15y$

Video Solution

VIDEO SOLUTION

Question 12

The minimum number of flowers that could be offered to each deity is:

- A 0
- B 15
- C 16
- D Cannot be determined

cracku

DIRECTIONS for the following questions: These questions are based on the situation given below: A young girl Roopa leaves home with x flowers, goes to the bank of a nearby river. On the bank of the river, there are four places of worship, standing in a row. She dips all the x flowers into the river. The number of flowers doubles. Then she enters the first place of worship, offers y flowers to the deity. She dips the remaining flowers into the river, and again the number of flowers doubles. She goes to the second place of worship, offers y flowers to the deity. She dips the remaining flowers into the river, and again the number of flowers doubles. She goes to the third place of worship, offers y flowers to the deity. She dips the remaining flowers into the river, and again the number of flowers doubles. She goes to the fourth place of worship, offers y flowers to the deity. Now she is left with no flowers in hand.

If Roopa leaves home with 30 flowers, the number of flowers she offers to each deity is:

A) 30
B) 32
C) 31
D) 33

Handwritten notes:

I. $2x - y$
II. $4x - 3y$
III. $8x - 6y$
IV. $16x - 15y$

Video Solution

VIDEO SOLUTION



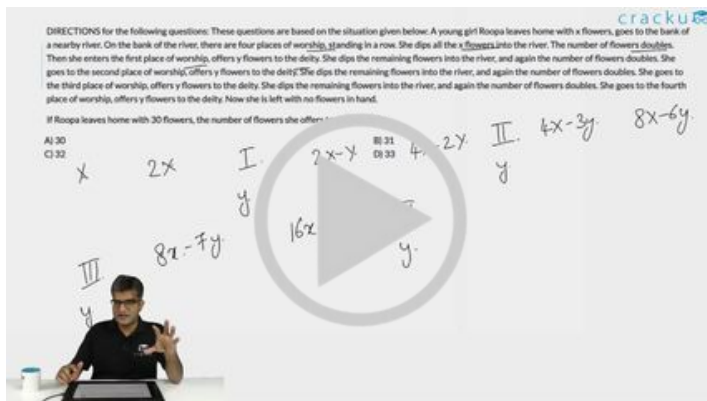
CAT Syllabus PDF



Question 13

If Roopa leaves home with 30 flowers, the number of flowers she offers to each deity is:

- A 30
- B 31
- C 32
- D 33

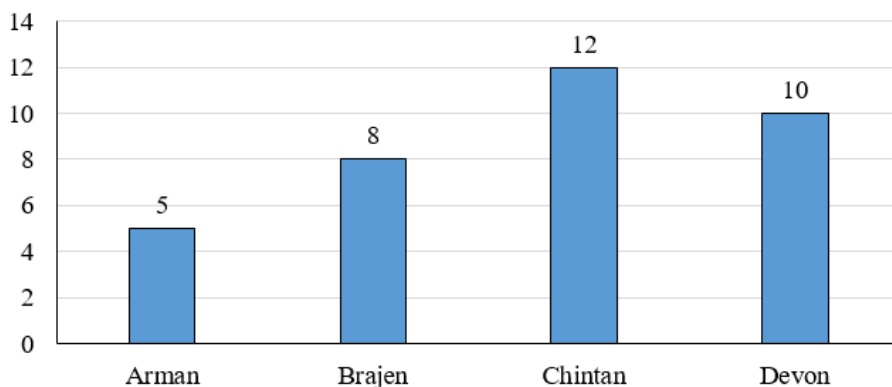


VIDEO SOLUTION

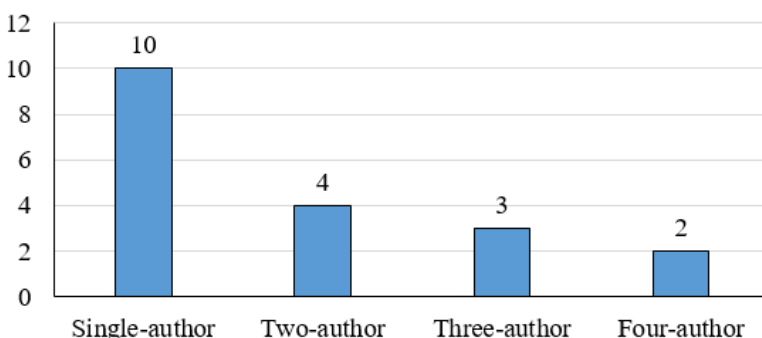
Instructions

The following charts depict details of research papers written by four authors, Arman, Brajen, Chintan, and Devon. The papers were of four types, single-author, two-author, three-author, and four-author, that is, written by one, two, three, or all four of these authors, respectively. No other authors were involved in writing these papers.

Number of papers by each author



Number of papers by type



The following additional facts are known.

- Each of the authors wrote at least one of each of the four types of papers.
- The four authors wrote different numbers of single-author papers.
- Both Chintan and Devon wrote more three-author papers than Brajen.
- The number of single-author and two-author papers written by Brajen were the same.

Question 14

What was the total number of two-author and three-author papers written by Brajen?

[VIEW SOLUTION](#)

Question 15

Which of the following statements is/are NECESSARILY true?

- i. Chintan wrote exactly three two-author papers.
- ii. Chintan wrote more single-author papers than Devon.

- A Neither i nor ii
- B Only i
- C Only ii
- D Both i and ii

[VIEW SOLUTION](#)



Question 16

Which of the following statements is/are NECESSARILY true?

- i. Arman wrote three-author papers only with Chintan and Devon.
- ii. Brajen wrote three-author papers only with Chintan and Devon.

- A Only ii
- B Both i and ii
- C Only i
- D Neither i or ii

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Question 17

If Devon wrote more than one two-author papers, then how many two-author papers did Chintan write?

[VIEW SOLUTION](#)

Instructions

All the first-year students in the computer science (CS) department in a university take both the courses (i) AI and (ii) ML. Students from other departments (non-CS students) can also take one of these two courses, but not both. Students who fail in a course get an F grade; others pass and are awarded A or B or C grades depending on their performance. The following are some additional facts about the number of students who took these two courses this year and the grades they obtained.

1. The numbers of non-CS students who took AI and ML were in the ratio 2 : 5.
2. The number of non-CS students who took either AI or ML was equal to the number of CS students.
3. The numbers of non-CS students who failed in the two courses were the same and their total is equal to the number of CS students who got a C grade in ML.
4. In both the courses, 50% of the students who passed got a B grade. But, while the numbers of students who got A and C grades were the same for AI, they were in the ratio 3 : 2 for ML.
5. No CS student failed in AI, while no non-CS student got an A grade in AI.
6. The numbers of CS students who got A, B and C grades respectively in AI were in the ratio 3 : 5 : 2, while in ML the ratio was 4 : 5 : 2.
7. The ratio of the total number of non-CS students failing in one of the two courses to the number of CS students failing in one of the two courses was 3 : 1.
8. 30 students failed in ML.

Question 18

How many students got A grade in AI?

- A 99
- B 42
- C 84
- D 63

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AI ML

A B C F A B C F

Non-CS (7x)

AI (2x) ML (5x)

A B C F A B C F

All the first-year students in the computer science (CS) department in a university take both the courses (i) AI and (ii) ML. Students from other departments (non-CS students) can also take one of these two courses, but not both. Students who fail in a course get an F grade; others pass and are awarded A or B or C grades depending on their performance. The following are some additional facts about the number of students who took these two courses this year and the grades they obtained.

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4. In both the courses, 50% of the students who passed got a B grade. But, while the numbers of students who got A and C grades were the same for AI, they were in the ratio 3 : 2 for ML.
5. No CS student failed in AI, while no non-CS student got an A grade in AI.
6. The numbers of CS students who got A, B and C grades respectively in AI were in the ratio 3 : 5 : 2, while in ML the ratio was 4 : 5 : 2.
7. The ratio of the total number of non-CS students failing in one of the two courses to the number of CS students failing in one of the two courses was 3 : 1.
8. 30 students failed in ML.

VIDEO SOLUTION



IIM Call Predictor

Accurate analysis of your profile



Question 19

How many CS students failed in ML?

 VIDEO SOLUTION

How many non-CS students got A grade in ML?

 VIDEO SOLUTION

How many students took AI?

- | | |
|----------|-----|
| A | 60 |
| B | 210 |
| C | 90 |
| D | 270 |

cracku

AI

ML

Non-CS (7X)

AI

ML

Non-CS (7X)

AI

ML

Non-CS (7X)

All the first-year students in the computer science (CS) department in a university take both the courses (i) AI and (ii) ML. Students from other departments (non-CS students) can also take one of these two courses, but not both. Students who fail in a course get an F grade; others pass and are awarded A or B or C grades depending on their performance. The following are some additional facts about the number of students who took these two courses this year and the grades they obtained.

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- The numbers of non-CS students who failed in the two courses were the same and the total is equal to the number of CS students who got a C grade in ML.
- In the courses, 50% of the students who passed got a B grade. But, while the A and C grades were the same for AI, they were in the ratio 3 : 5 : 2, while in ML, the ratio was 4 : 5 : 2.
- The ratio of the total number of non-CS students failing in one of the two courses to the number of CS students failing in one of the two courses was 3 : 1.
- 30 students failed in ML.

VIDEO SOLUTION

CAT Percentile Predictor



Question 22

How many non-CS students got B grade in ML?

- A 165
- B 75
- C 25
- D 90

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AI

ML

Non-CS (7X)

AI

ML

Non-CS (7X)

AI

ML

Non-CS (7X)

All the first-year students in the computer science (CS) department in a university take both the courses (i) AI and (ii) ML. Students from other departments (non-CS students) can also take one of these two courses, but not both. Students who fail in a course get an F grade; others pass and are awarded A or B or C grades depending on their performance. The following are some additional facts about the number of students who took these two courses this year and the grades they obtained.

- The number of non-CS students who took AI and ML were in the ratio 2 : 5.
- The number of CS students who took either AI or ML was equal to the number of CS students.
- The numbers of non-CS students who failed in the two courses were the same and the total is equal to the number of CS students who got a C grade in ML.
- In the courses, 50% of the students who passed got a B grade. But, while the A and C grades were the same for AI, they were in the ratio 3 : 5 : 2, while in ML, the ratio was 4 : 5 : 2.
- The ratio of the total number of non-CS students failing in one of the two courses to the number of CS students failing in one of the two courses was 3 : 1.
- 30 students failed in ML.

VIDEO SOLUTION

Instructions

Three participants - Akhil, Bimal and Chatur participate in a random draw competition for five days. Every day, each participant randomly picks up a ball numbered between 1 and 9. The number on the ball determines his score on that day. The total score of a participant is the sum of his scores attained in the five days. The total score of a day is the sum of participants' scores on that day. The 2-day average on a day, except on Day 1, is the average of the total scores of that day and of the previous day. For example, if the total scores of Day 1 and Day 2 are 25 and 20, then the 2-day average on Day 2 is calculated as 22.5. Table 1 gives the 2-day averages for Days 2 through 5.

Table 1: 2-day averages for Days 2 through 5			
Day 2	Day 3	Day 4	Day 5
15	15.5	16	17

Participants are ranked each day, with the person having the maximum score being awarded the minimum rank (1) on that day. If there is a tie, all participants with the tied score are awarded the best available rank. For example, if on a day Akhil, Bimal, and Chatur score 8, 7 and 7 respectively, then their ranks will be 1, 2 and 2 respectively on that day. These ranks are given in Table 2.

Table 2: Ranks of participants on each day					
	Day 1	Day 2	Day 3	Day 4	Day 5
Akhil	1	2	2	3	3
Bimal	2	3	2	1	1
Chatur	3	1	1	2	2

The following information is also known.

1. Chatur always scores in multiples of 3. His score on Day 2 is the unique highest score in the competition. His minimum score is observed only on Day 1, and it matches Akhil's score on Day 4.
2. The total score on Day 3 is the same as the total score on Day 4.
3. Bimal's scores are the same on Day 1 and Day 3.

Question 23

If Akhil attains a total score of 24, then what is the total score of Bimal?

Three participants - Akhil, Bimal and Chatur participate in a random-draw competition for five days. Every day each participant randomly picks up a ball numbered between 1 and 9. The number on the ball determines his score on that day. The total score of a participant is the sum of his scores attained in the five days. The total score of a day is the sum of participants' scores on that day. The 2-day average on a day except on Day 1, is the average of the total scores of that day and of the 2-day average of the previous day. For example, if the total scores of Day 1 and Day 2 are 25 and 20, then the 2-day average on Day 3 is calculated as 22.5. Table 1 gives the 2-day averages for Days 2 through 5.

Table 1: 2-day averages for Days 2 through 5

Day 2	Day 3	Day 4	Day 5
15	15.5	16	17

Each day, participants are ranked. The person having the maximum score being awarded the minimum rank (1) on that day. If there is a tie, all participants with the tied score are awarded the best available rank. For example, if on a day Akhil, Bimal, and Chatur score 8, 7 and 7 respectively, then their ranks will be 1, 2 and 2 respectively on that day. These ranks are given in Table 2.

Table 2: Ranks of participants on each day

	Day 1	Day 2	Day 3	Day 4	Day 5
Akhil	1	2	2	3	3
Bimal	2	3	2	1	1
Chatur	3	1	1	2	2

The following information is also known.

1. Chatur always scores in multiples of 3. His score on Day 2 is the unique highest score in the competition. His minimum score is observed only on Day 1, and it matches Akhil's score on Day 4.
2. The total score on Day 3 is the same as the total score on Day 4.
3. Bimal's scores are the same on Day 1 and Day 3.

VIDEO SOLUTION

Question 24

What is the minimum possible total score of Bimal?

Three participants - Akhil, Bimal and Chatur participate in a random-draw competition for five days. Every day each participant randomly picks up a ball numbered between 1 and 9. The number on the ball determines his score on that day. The total score of a participant is the sum of his scores attained in the five days. The total score of a day is the sum of participants' scores on that day. The 2-day average on a day except on Day 1, is the average of the total scores of that day and of the 2-day average of the previous day. For example, if the total scores of Day 1 and Day 2 are 25 and 20, then the 2-day average on Day 3 is calculated as 22.5. Table 1 gives the 2-day averages for Days 2 through 5.

Table 1: 2-day averages for Days 2 through 5

Day 2	Day 3	Day 4	Day 5
15	15.5	16	17

Each day, participants are ranked. The person having the maximum score being awarded the minimum rank (1) on that day. If there is a tie, all participants with the tied score are awarded the best available rank. For example, if on a day Akhil, Bimal, and Chatur score 8, 7 and 7 respectively, then their ranks will be 1, 2 and 2 respectively on that day. These ranks are given in Table 2.

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	Day 1	Day 2	Day 3	Day 4	Day 5
Akhil	1	2	2	3	3
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Chatur	3	1	1	2	2

The following information is also known.

1. Chatur always scores in multiples of 3. His score on Day 2 is the unique highest score in the competition. His minimum score is observed only on Day 1, and it matches Akhil's score on Day 4.
2. The total score on Day 3 is the same as the total score on Day 4.
3. Bimal's scores are the same on Day 1 and Day 3.

VIDEO SOLUTION

Question 25

If the total score of Bimal is a multiple of 3, what is the score of Akhil on Day 2?

- A Cannot be determined
- B 5
- C 6
- D 4

	D1	D2	D3	D4	D5
A					
B					
C					

Three participants - Akhil, Bimal and Chatur participate in a random draw competition for five days. Every day, each participant randomly picks up a ball numbered between 1 and 9. The number on the ball determines his score on that day. The total score of a participant is the sum of his scores attained in the five days. The total score of a day is the sum of participants' scores on that day. The 2-day average on a day, except on Day 1, is the average of the total scores of that day and of the day before it. For example, if the total scores of Day 1 and Day 2 are 25 and 20, then the 2-day average on Day 2 is calculated as 22.5. Table 1 gives the 2-day averages for Days 2 through 5.

Day	Day 2	Day 3	Day 4	Day 5
Average	22.5	20	25	22.5

The following information is also known:

- Chatur always scores in multiples of 3. His score on Day 2 is the minimum score in the competition. His minimum score is observed only on Day 2.
- The total score on Day 3 is the same as the total score on Day 4.

VIDEO SOLUTION

Question 26

What is Akhil's score on Day 1?

- A 5
- B 7
- C 6
- D 8

	D1	D2	D3	D4	D5
A					
B					
C					

Three participants - Akhil, Bimal and Chatur participate in a random draw competition for five days. Every day, each participant randomly picks up a ball numbered between 1 and 9. The number on the ball determines his score on that day. The total score of a participant is the sum of his scores attained in the five days. The total score of a day is the sum of participants' scores on that day. The 2-day average on a day, except on Day 1, is the average of the total scores of that day and of the day before it. For example, if the total scores of Day 1 and Day 2 are 25 and 20, then the 2-day average on Day 2 is calculated as 22.5. Table 1 gives the 2-day averages for Days 2 through 5.

Day	Day 2	Day 3	Day 4	Day 5
Average	22.5	20	25	22.5

The following information is also known:

- Chatur always scores in multiples of 3. His score on Day 2 is the minimum score in the competition. His minimum score is observed only on Day 2.
- The total score on Day 3 is the same as the total score on Day 4.

VIDEO SOLUTION

Question 27

Who attains the maximum total score?

- A Cannot be determined
- B Akhil
- C Bimal
- D Chatur

Three participants - Akhil, Bimal and Chatur participate in a random-draw competition for five days. Every day, each participant randomly picks up a ball numbered between 1 and 9. The number on the ball determines his score on that day. The total score of a participant is the sum of his scores attained in the five days. The total score of a day is the sum of participants' scores on that day. The 2-day average on a day, except on Day 1, is the average of the total scores of that day and of the previous day. For example, if the total scores of Day 1 and Day 2 are 25 and 20, then the 2-day average on Day 2 is calculated as 22.5. Table 1 gives the 2-day averages for Days 2 through 5.

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Bimal					
Chatur					

Table 1: 2-day averages for Days 2 through 5

Day	Day 2	Day 3	Day 4	Day 5
Average	22.5	23.5	24.5	25.5

The following information is also known:

- Chatur always scores in multiples of 3. His score on Day 2 is the maximum score in the competition. His minimum score is observed only on one day.
- The total score on Day 3 is the same as the total score on Day 4.

VIDEO SOLUTION

CAT Crash Course

Live Classes by IIMA alumni

Instructions

Aurevia, Brelosia, Cyrenia and Zerathania are four countries with their currencies being Aurels, Brins, Crowns, and Zentars, respectively. The currencies have different exchange values. Crown's currency exchange rate with Zentars = 0.5, i.e., 1 Crown is worth 0.5 Zentars.

Three travelers, Jano, Kira, and Lian set out from Zerathania visiting exactly two of the countries. Each country is visited by exactly two travelers. Each traveler has a unique Flight Cost, which represents the total cost of airfare in traveling to both the countries and back to Zerathania. The Flight Cost of Jano was 4000 Zentars, while that of the other two travelers were 5000 and 6000 Zentars, not necessarily in that order. When visiting a country, a traveler spent either 1000, 2000 or 3000 in the country's local currency. Each traveler had different spends (in the country's local currency) in the two countries he/she visited. Across all the visits, there were exactly two spends of 1000 and exactly one spend of 3000 (in the country's local currency).

The total "Travel Cost" for a traveler is the sum of his/her Flight Cost and the money spent in the countries visited.

The citizens of the four countries with knowledge of these travels made a few observations, with spends measured in their respective local currencies:

- Aurevia citizen: Jano and Kira visited our country, and their Travel Costs were 3500 and 8000, respectively.
- Brelosia citizen: Kira and Lian visited our country, spending 2000 and 3000, respectively. Kira's Travel Cost was 4000.
- Cyrenia citizen: Lian visited our country and her Travel Cost was 36000.

Question 28

What was Jano's total spend in the two countries he visited, in Aurels?

[VIEW SOLUTION](#)

Question 29

One Brin is equivalent to how many Crowns?

- A 8
- B 0.125
- C 0.5
- D 4

[VIEW SOLUTION](#)

Question 30

How many Zentars did Lian spend in the two countries he visited?

[VIEW SOLUTION](#)



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Question 31

Which of the following statements is NOT true about money spent in the local currency?

- A Jano spent 2000 in Aurevia
- B Lian spent 2000 in Cyrenia
- C Jano spent 2000 in Cyrenia
- D Kira spent 1000 in Aurevia

[VIEW SOLUTION](#)

Question 32

What is the sum of Travel Costs for all travelers in Zentars?

[VIEW SOLUTION](#)

Instructions

An air conditioner (AC) company has four dealers - D1, D2, D3 and D4 in a city. It is evaluating sales performances of these dealers. The company sells two variants of ACs - Window and Split. Both these variants can be either Inverter type or Non-inverter type. It is known that of the total number of ACs sold in the city, 25% were of Window variant, while the rest were of Split variant. Among the Inverter ACs sold, 20% were of Window variant.

The following information is also known:

1. Every dealer sold at least two window ACs.
2. D1 sold 13 inverter ACs, while D3 sold 5 Non-inverter ACs.
3. A total of six Window Non-inverter ACs and 36 Split Inverter ACs were sold in the city.
4. The number of Split ACs sold by D1 was twice the number of Window ACs sold by it.

5. D3 and D4 sold an equal number of Window ACs and this number was one-third of the number of similar ACs sold by D2.
6. D2 and D3 were the only ones who sold Window Non-inverter ACs. The number of these ACs sold by D2 was twice the number of these ACs sold by D3.
7. D3 and D4 sold an equal number of Split Inverter ACs. This number was half the number of similar ACs sold by D2.

Question 33

How many Split Inverter ACs did D2 sell?

	Window		Split	
	I-NV	N-I-NV	I-NV	N-I-NV
D ₁	P		13-P	
D ₂				
D ₃		Q		5-Q
D ₄				
Total				

An air conditioner (AC) company has four dealers - D₁, D₂, D₃ and D₄ in a city. It is evaluating sales performances of these dealers. The company sells two variants of ACs - Window and Split. Both these variants can be either Inverter type or Non-Inverter type. It is known that of the total number of ACs sold in the city, 25% were of Window variant, while the rest were of Split variant. Among the Inverter ACs sold, 20% were of Window variant.

Information is also known:

- Every dealer sold at least two window ACs.
- D₁ sold 13 Inverter ACs, while D₃ sold 5 Non-Inverter ACs.
- A total of six Window Non-Inverter ACs and 36 Split Inverter ACs were sold in the city.

Number of Split ACs sold by D₁ was twice the number of Window ACs sold by D₁.

Number of Window ACs sold by D₂ was one-third of the number of Window ACs sold by D₃.

Number of Window Non-Inverter ACs sold by D₂ was twice the number of these ACs sold by D₃.

D₃ and D₄ sold an equal number of Split Inverter ACs. This number was half the number of similar ACs sold by D₂.

VIDEO SOLUTION

CAT Daily Targets

Step by step, heading towards 99+ percentile

Question 34

Which of the following statements is necessarily false?

- A D2 sold the highest number of ACs.
- B D4 sold more Split ACs as compared to D3.
- C D1 and D3 sold an equal number of Split ACs.
- D D1 and D3 together sold more ACs as compared to D2 and D4 together.

	Window		Split	
	I-NV	N-I-NV	I-NV	N-I-NV
D ₁	P		13-P	
D ₂				
D ₃		Q		5-Q
D ₄				
Total				

An air conditioner (AC) company has four dealers - D₁, D₂, D₃ and D₄ in a city. It is evaluating sales performances of these dealers. The company sells two variants of ACs - Window and Split. Both these variants can be either Inverter type or Non-Inverter type. It is known that of the total number of ACs sold in the city, 25% were of Window variant, while the rest were of Split variant. Among the Inverter ACs sold, 20% were of Window variant.

Information is also known:

- Every dealer sold at least two window ACs.
- D₁ sold 13 Inverter ACs, while D₃ sold 5 Non-Inverter ACs.
- A total of six Window Non-Inverter ACs and 36 Split Inverter ACs were sold in the city.

Number of Split ACs sold by D₁ was twice the number of Window ACs sold by D₁.

Number of Window ACs sold by D₂ was one-third of the number of Window ACs sold by D₃.

Number of Window Non-Inverter ACs sold by D₂ was twice the number of these ACs sold by D₃.

D₃ and D₄ sold an equal number of Split Inverter ACs. This number was half the number of similar ACs sold by D₂.

VIDEO SOLUTION

Question 35

If D3 and D4 sold an equal number of ACs, then what was the number of Non-inverter ACs sold by D2?

- A 4
- B 5
- C 7
- D 6

The video solution shows a table with columns for Dealer (D1, D2, D3, D4, Total) and rows for AC types (Window, Split, Inverter, Non-Inverter). The table is partially filled with handwritten values. The text explains the problem and provides a step-by-step solution.

	Window	Split	Inverter	Non-Inverter
D1	10	10	10	10
D2				
D3		10		5-10
D4				
Total				

Handwritten notes on the video include: "D1 sold 13 Inverter ACs, while D3 sold 5 Non-Inverter ACs." and "D3 and D4 sold an equal number of ACs, then what was the number of Non-Inverter ACs sold by D2?"

[VIDEO SOLUTION](#)

Instructions

DIRECTIONS for the following questions: These questions are based on the situation given below:

There are m blue vessels with known volumes $V_1, V_2, \dots, V_m$, arranged in ascending order of volume, where $v_1 > 0.5$ litre, and $V_m < 1$ litre. Each of these is full of water initially. The water from each of these is emptied into a minimum number of empty white vessels, each having volume 1 litre. The water from a blue vessel is not emptied into a white vessel unless the white vessel has enough empty volume to hold all the water of the blue vessel. The number of white vessels required to empty all the blue vessels according to the above rules was n .

Question 36

Among the four values given below, which is the least upper bound on e , where e is the total empty volume in the m white vessels at the end of the above process?

- A mv_m
- B $m(1 - v_m)$
- C mv_1
- D $m(1 - v_1)$

[VIEW SOLUTION](#)

Advertisement for CAT Sectional Tests. It features a person sitting at a desk with a laptop, a "FREE" tag, and the text "CAT Sectional Tests". There is also a "Take Here" button.

Question 37

Let the number of white vessels needed be n_1 for the emptying process described above, if the volume of each white vessel is 2 liters. Among the following values, which is the least upper bound on n_1 ?

- A** $m/4$
- B** smallest integer greater than or equal to $(m/2)$
- C** n
- D** greatest integer less than or equal to $(m/2)$

 [VIEW SOLUTION](#)

Answers

1.C	2.4	3.9	4.D	5.5	6.50	7.B	8.80
9.40	10.60	11.B	12.C	13.C	14.4	15.A	16.B
17.3	18.D	19.12	20.27	21.D	22.B	23.26	24.25
25.D	26.B	27.D	28.1500	29.A	30.13000	31.A	32.41000
33.14	34.D	35.B	36.B	37.B			

Explanations

1. C

We are given that each box contains three sacks. Each sack has a certain number of coins, between 1 and 9, both inclusive.

The average number of coins per sack in the boxes are all distinct integers. The total number of coins in each row is the same. The total number of coins in each column is also the same.

=> The total number of coins in a box range from 3 ($1 + 1 + 1$) to 27 ($9 + 9 + 9$)

Since, it is given that the average number of coins per sack in the boxes are all distinct integers => The total number of coins in a box would be 3, 6, 9, 12, 15, 18, 21, 24, 27 => averages of 1, 2, 3, 4, ..., 9 => Sum = 45.

=> Sum of averages coins in a box in a row or column = $45/3 = 15$ [The total number of coins in each row is the same. The total number of coins in each column is also the same.] ==> (1)

Let us represent the final configuration of the sacks in boxes as follows:

		Table	
	C-1	C-2	C-3
R-1			
R-2			
R-3			

Also a bag (x,y) => bag in xth row and yth column.

We are given 2 clues => Table-1 & Table-2

Consider bag (3,1)

=> From Table-1 => Median = 8 & From Table-2 all 3 sacks have more than 5 coins. Also * => There is a 9 in one of the sacks.

=> c, 8, 9 are the coins in bag (3,1), now $c > 5$ & $c + 8 + 9$ should be a multiple of 3 => $c = 7$ is the only possibility.

=> bag (3,1) has 7, 8, 9 coins with average = 8.

Consider bag (2,1)

Median = 2 and 1 sack has more than 5 coins. Also ** => conditions i & iii should be satisfied.

=> 1, 2, 9 are the coins in bag (2,1) with average = 4

Consider bag (1,2)

Median = 9 and 2 elements are more than 5. Also * => (9 is present & 1 is not present)

=> c, 9, 9 are the coins in bag (1,2) and c is not equal to 1 and less than 5 => $c = 3$ for $c + 18$ to be a multiple of 3.

=> 3, 9, 9 are the coins in bag (1,2) with average = 7.

Capturing this info. in the table:

		Table	
	C-1	C-2	C-3
R-1		3,9,9 (7)	
R-2	1,2,9 (4)		
R-3	7,8,9 (8)		

From (1), The average in bag (1,1) is $15 - 4 - 8 = 3$.

From (1), The average in bag (1,3) is $15 - 3 - 7 = 5$.

		Table	
	C-1	C-2	C-3
R-1	Avg = 3	3,9,9 (7)	Avg = 5
R-2	1,2,9 (4)		
R-3	7,8,9 (8)		

Consider bag (1,1)

Avg = 3, 1 sack has more than 5 and ** $\Rightarrow$ 2 conditions are being satisfied. $\Rightarrow$ (can't be condition-3 $\Rightarrow$ 9 coins as the total sum of coins is it self $3 \times 3 = 9$)

$\Rightarrow$ bag (1,1) has 1, 1, 7 coins with average = 3.

Consider bag (1,3)

Avg. = 5 $\Rightarrow$ Sum = 15.

Median = 6 and 2 sacks have more than 5 and * $\Rightarrow$ (1 condition is satisfied)

Not condition ii as the median is 6 & Not condition iii as the sum of 2 sacks itself will become $6 + 9 = 15$

$\Rightarrow$ 1, 6, c are the coins $\Rightarrow$ For sum = 15 $\Rightarrow c = 15 - 1 - 6 = 8$

$\Rightarrow$ bag (1,3) has 1, 6, 8 coins with average = 5.

		Table	
	C-1	C-2	C-3
R-1	1,1,7 (3)	3,9,9 (7)	1,6,8 (5)
R-2	1,2,9 (4)		
R-3	7,8,9 (8)		

Consider bag (3,3)

0 sacks have more than 5 coins and ** $\Rightarrow$ conditions i & ii are being satisfied.

$\Rightarrow$ 1,1,c are the coins. Now $c = 1$ or 2 or 3 or 4 $\Rightarrow c = 1$ or 4 for number of coins to be a multiple of 3.

But $c = 1$ as no other bag has the possibility to get avg. = 1 as bag (2,2) should have 1, b, c coins and b and c should be more than 1 as only 1^*

$\Rightarrow$ bag (3,3) has 1, 1, 1 coins with average = 1.

Now, we can fill the averages in all the bags.

		Table	
	C-1	C-2	C-3
R-1	1,1,7 (3)	3,9,9 (7)	1,6,8 (5)
R-2	1,2,9 (4)	Avg = 2	Avg = 9
R-3	7,8,9 (8)	Avg = 6	1,1,1 (1)

In bag (2,3) Avg. = 9 $\Rightarrow$ 9, 9, 9 are the coins.

In bag (2,2) $\Rightarrow$ Avg. = 2 $\Rightarrow$ Sum = 6 and only 1^* $\Rightarrow$ smallest element should be 1.

$\Rightarrow$ 1, b, c are the coins where $b + c = 5$ and b,c can't be equal to 1 and less than 5 $\Rightarrow 2 + 3 = 5$ is the only possibility.

$\Rightarrow$ 1, 2, 3 are the coins with average = 2.

	Table		
	C-1	C-2	C-3
R-1	1,1,7 (3)	3,9,9 (7)	1,6,8 (5)
R-2	1,2,9 (4)	1,2,3 (2)	9,9,9 (9)
R-3	7,8,9 (8)	Avg = 6	1,1,1 (1)

Considering bag (3,2)

Avg. = 6 $\Rightarrow$ Sum = 18.

2 sacks more than 5 coins and $** \Rightarrow$ 2 sacks have 1 and 9 coins.

$\Rightarrow$ bag (3,2) has 1, c, 9 coins and $c = 18 - 1 - 9 = 8$

$\Rightarrow$ bag (3,2) has 1, 8, 9 coins with average = 6 coins.

$\Rightarrow$ Final required table, bracket number $\Rightarrow$ average coins per sack in the bag.

	Table		
	C-1	C-2	C-3
R-1	1,1,7 (3)	3,9,9 (7)	1,6,8 (5)
R-2	1,2,9 (4)	1,2,3 (2)	9,9,9 (9)
R-3	7,8,9 (8)	1,8,9 (6)	1,1,1 (1)

Bags (2,1), (3,1), (1,2), (3,2), (2,3) have at least 1 sack with 9 coins. $\Rightarrow$ Total of 5 bags.

 VIDEO SOLUTION

2.4

We are given that each box contains three sacks. Each sack has a certain number of coins, between 1 and 9, both inclusive.

The average number of coins per sack in the boxes are all distinct integers. The total number of coins in each row is the same. The total number of coins in each column is also the same.

$\Rightarrow$ The total number of coins in a box range from 3 ($1 + 1 + 1$) to 27 ($9 + 9 + 9$)

Since, it is given that the average number of coins per sack in the boxes are all distinct integers $\Rightarrow$ The total number of coins in a box would be 3, 6, 9, 12, 15, 18, 21, 24, 27 $\Rightarrow$ averages of 1, 2, 3, 4, ..., 9 $\Rightarrow$ Sum = 45.

$\Rightarrow$ Sum of averages coins in a box in a row or column = $45/3 = 15$ [The total number of coins in each row is the same. The total number of coins in each column is also the same.] $\Rightarrow$ (1)

Let us represent the final configuration of the sacks in boxes as follows:

	Table		
	C-1	C-2	C-3
R-1			
R-2			
R-3			

Also a bag (x,y) $\Rightarrow$ bag in xth row and yth column.

We are given 2 clues $\Rightarrow$ Table-1 & Table-2

Consider bag (3,1)

$\Rightarrow$ From Table-1 $\Rightarrow$ Median = 8 & From Table-2 all 3 sacks have more than 5 coins. Also $*$ $\Rightarrow$ There is a 9 in one of the sacks.

$\Rightarrow$ c, 8, 9 are the coins in bag (3,1), now $c > 5$ & $c + 8 + 9$ should be a multiple of 3 $\Rightarrow c = 7$ is the only possibility.

$\Rightarrow$ bag (3,1) has 7, 8, 9 coins with average = 8.

Consider bag (2,1)

Median = 2 and 1 sack has more than 5 coins. Also $** \Rightarrow$ conditions i & iii should be satisfied.

$\Rightarrow$ 1, 2, 9 are the coins in bag (2,1) with average = 4

Consider bag (1,2)

Median = 9 and 2 elements are more than 5. Also * $\Rightarrow$ (9 is present & 1 is not present)

$\Rightarrow$ c, 9, 9 are the coins in bag (1,2) and c is not equal to 1 and less than 5 $\Rightarrow$ c = 3 for c + 18 to be a multiple of 3.

$\Rightarrow$ 3, 9, 9 are the coins in bag (1,2) with average = 7.

Capturing this info. in the table:

		Table	
	C-1	C-2	C-3
R-1		3,9,9 (7)	
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R-3	7,8,9 (8)		

From (1), The average in bag (1,1) is $15 - 4 - 8 = 3$.

From (1), The average in bag (1,3) is $15 - 3 - 7 = 5$.

		Table	
	C-1	C-2	C-3
R-1	Avg = 3	3,9,9 (7)	Avg = 5
R-2	1,2,9 (4)		
R-3	7,8,9 (8)		

Consider bag (1,1)

Avg = 3, 1 sack has more than 5 and ** $\Rightarrow$ 2 conditions are being satisfied. $\Rightarrow$ (can't be condition-3 $\Rightarrow$ 9 coins as the total sum of coins is it self $3 \times 3 = 9$)

$\Rightarrow$ bag (1,1) has 1, 1, 7 coins with average = 3.

Consider bag (1,3)

Avg. = 5 $\Rightarrow$ Sum = 15.

Median = 6 and 2 sacks have more than 5 and * $\Rightarrow$ (1 condition is satisfied)

Not condition ii as the median is 6 & Not condition iii as the sum of 2 sacks itself will become $6 + 9 = 15$

$\Rightarrow$ 1, 6, c are the coins $\Rightarrow$ For sum = 15 $\Rightarrow$ c = $15 - 1 - 6 = 8$

$\Rightarrow$ bag (1,3) has 1, 6, 8 coins with average = 5.

		Table	
	C-1	C-2	C-3
R-1	1,1,7 (3)	3,9,9 (7)	1,6,8 (5)
R-2	1,2,9 (4)		
R-3	7,8,9 (8)		

Consider bag (3,3)

0 sacks have more than 5 coins and ** $\Rightarrow$ conditions i & ii are being satisfied.

$\Rightarrow$ 1,1,c are the coins. Now c = 1 or 2 or 3 or 4 $\Rightarrow$ c = 1 or 4 for number of coins to be a multiple of 3.

But c = 1 as no other bag has the possibility to get avg. = 1 as bag (2,2) should have 1, b, c coins and b and c should be more than 1 as only 1*

$\Rightarrow$ bag (3,3) has 1, 1, 1 coins with average = 1.

Now, we can fill the averages in all the bags.

		Table	
	C-1	C-2	C-3
R-1	1,1,7 (3)	3,9,9 (7)	1,6,8 (5)
R-2	1,2,9 (4)	Avg = 2	Avg = 9
R-3	7,8,9 (8)	Avg = 6	1,1,1 (1)

In bag (2,3) Avg. = 9 $\Rightarrow$ 9, 9, 9 are the coins.

In bag (2,2) => Avg. = 2 => Sum = 6 and only 1* => smallest elements should be 1.

=> 1, b, c are the coins where $b + c = 5$ and b, c can't be equal to 1 and less than 5 => $2 + 3 = 5$ is the only possibility.

=> 1, 2, 3 are the coins with average = 2.

	Table		
	C-1	C-2	C-3
R-1	1,1,7 (3)	3,9,9 (7)	1,6,8 (5)
R-2	1,2,9 (4)	1,2,3 (2)	9,9,9 (9)
R-3	7,8,9 (8)	Avg = 6	1,1,1 (1)

Considering bag (3,2)

Avg. = 6 => Sum = 18.

2 sacks more than 5 coins and ** => 2 sacks have 1 and 9 coins.

=> bag (3,2) has 1, c, 9 coins and $c = 18 - 1 - 9 = 8$

=> bag (3,2) has 1, 8, 9 coins with average = 6 coins.

=> Final required table, bracket number => average coins per sack in the bag.

	Table		
	C-1	C-2	C-3
R-1	1,1,7 (3)	3,9,9 (7)	1,6,8 (5)
R-2	1,2,9 (4)	1,2,3 (2)	9,9,9 (9)
R-3	7,8,9 (8)	1,8,9 (6)	1,1,1 (1)

Average = Median in boxes (3,1), (2,2), (2,3) and (3,3) => 4 boxes.

VIDEO SOLUTION

3.9

We are given that each box contains three sacks. Each sack has a certain number of coins, between 1 and 9, both inclusive.

The average number of coins per sack in the boxes are all distinct integers. The total number of coins in each row is the same. The total number of coins in each column is also the same.

=> The total number of coins in a box range from 3 ($1 + 1 + 1$) to 27 ($9 + 9 + 9$)

Since, it is given that the average number of coins per sack in the boxes are all distinct integers => The total number of coins in a box would be 3, 6, 9, 12, 15, 18, 21, 24, 27 => averages of 1, 2, 3, 4, ..., 9 => Sum = 45.

=> Sum of averages coins in a box in a row or column = $45/3 = 15$ [The total number of coins in each row is the same. The total number of coins in each column is also the same.] => (1)

Let us represent the final configuration of the sacks in boxes as follows:

	Table		
	C-1	C-2	C-3
R-1			
R-2			
R-3			

Also a bag (x,y) => bag in xth row and yth column.

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=> c, 8, 9 are the coins in bag (3,1), now $c > 5$ & $c + 8 + 9$ should be a multiple of 3 => $c = 7$ is the only possibility.

=> bag (3,1) has 7, 8, 9 coins with average = 8.

Consider bag (2,1)

Median = 2 and 1 sack has more than 5 coins. Also ** => conditions i & iii should be satisfied.

=> 1, 2, 9 are the coins in bag (2,1) with average = 4

Consider bag (1,2)

Median = 9 and 2 elements are more than 5. Also * => (9 is present & 1 is not present)

=> c, 9, 9 are the coins in bag (1,2) and c is not equal to 1 and less than 5 => c = 3 for c + 18 to be a multiple of 3.

=> 3, 9, 9 are the coins in bag (1,2) with average = 7.

Capturing this info. in the table:

		Table	
	C-1	C-2	C-3
R-1		3,9,9 (7)	
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R-3	7,8,9 (8)		

From (1), The average in bag (1,1) is $15 - 4 - 8 = 3$.

From (1), The average in bag (1,3) is $15 - 3 - 7 = 5$.

		Table	
	C-1	C-2	C-3
R-1	Avg = 3	3,9,9 (7)	Avg = 5
R-2	1,2,9 (4)		
R-3	7,8,9 (8)		

Consider bag (1,1)

Avg = 3, 1 sack has more than 5 and ** => 2 conditions are being satisfied. => (can't be condition-3 => 9 coins as the total sum of coins is it self $3*3 = 9$)

=> bag (1,1) has 1, 1, 7 coins with average = 3.

Consider bag (1,3)

Avg. = 5 => Sum = 15.

Median = 6 and 2 sacks have more than 5 and * => (1 condition is satisfied)

Not condition ii as the median is 6 & Not condition iii as the sum of 2 sacks itself will become $6 + 9 = 15$

=> 1, 6, c are the coins => For sum = 15 => $c = 15 - 1 - 6 = 8$

=> bag (1,3) has 1, 6, 8 coins with average = 5.

		Table	
	C-1	C-2	C-3
R-1	1,1,7 (3)	3,9,9 (7)	1,6,8 (5)
R-2	1,2,9 (4)		
R-3	7,8,9 (8)		

Consider bag (3,3)

0 sacks have more than 5 coins and ** => conditions i & ii are being satisfied.

=> 1,1,c are the coins. Now $c = 1$ or 2 or 3 or 4 => $c = 1$ or 4 for number of coins to be a multiple of 3.

But $c = 1$ as no other bag has the possibility to get avg. = 1 as bag (2,2) should have 1, b, c coins and b and c should be more than 1 as only 1*

=> bag (3,3) has 1, 1, 1 coins with average = 1.

Now, we can fill the averages in all the bags.

	Table		
	C-1	C-2	C-3
R-1	1,1,7 (3)	3,9,9 (7)	1,6,8 (5)
R-2	1,2,9 (4)	Avg = 2	Avg = 9
R-3	7,8,9 (8)	Avg = 6	1,1,1 (1)

In bag (2,3) Avg. = 9 $\Rightarrow$ 9, 9, 9 are the coins.

In bag (2,2) $\Rightarrow$ Avg. = 2 $\Rightarrow$ Sum = 6 and only 1* $\Rightarrow$ smallest element should be 1.

$\Rightarrow$ 1, b, c are the coins where $b + c = 5$ and b,c can't be equal to 1 and less than 5 $\Rightarrow 2 + 3 = 5$ is the only possibility.

$\Rightarrow$ 1, 2, 3 are the coins with average = 2.

	Table		
	C-1	C-2	C-3
R-1	1,1,7 (3)	3,9,9 (7)	1,6,8 (5)
R-2	1,2,9 (4)	1,2,3 (2)	9,9,9 (9)
R-3	7,8,9 (8)	Avg = 6	1,1,1 (1)

Considering bag (3,2)

Avg. = 6 $\Rightarrow$ Sum = 18.

2 sacks more than 5 coins and ** $\Rightarrow$ 2 sacks have 1 and 9 coins.

$\Rightarrow$ bag (3,2) has 1, c, 9 coins and $c = 18 - 1 - 9 = 8$

$\Rightarrow$ bag (3,2) has 1, 8, 9 coins with average = 6 coins.

$\Rightarrow$ Final required table, bracket number $\Rightarrow$ average coins per sack in the bag.

	Table		
	C-1	C-2	C-3
R-1	1,1,7 (3)	3,9,9 (7)	1,6,8 (5)
R-2	1,2,9 (4)	1,2,3 (2)	9,9,9 (9)
R-3	7,8,9 (8)	1,8,9 (6)	1,1,1 (1)

Bag (1,1) $\Rightarrow$ 2 sacks with 1 coins, (2,1) $\Rightarrow$ 1 sack, (2,2) $\Rightarrow$ 1 sack, (3,2) $\Rightarrow$ 1 sack, (1,3) $\Rightarrow$ 1 sack, (3,3) $\Rightarrow$ 3 sacks.

$\Rightarrow$ Total = $2 + 1 + 1 + 1 + 1 + 3 = 9$ sacks.

 VIDEO SOLUTION

4. D

We are given that each box contains three sacks. Each sack has a certain number of coins, between 1 and 9, both inclusive.

The average number of coins per sack in the boxes are all distinct integers. The total number of coins in each row is the same. The total number of coins in each column is also the same.

$\Rightarrow$ The total number of coins in a box range from 3 ($1 + 1 + 1$) to 27 ($9 + 9 + 9$)

Since, it is given that the average number of coins per sack in the boxes are all distinct integers $\Rightarrow$ The total number of coins in a box would be 3, 6, 9, 12, 15, 18, 21, 24, 27 $\Rightarrow$ averages of 1, 2, 3, 4, ..., 9 $\Rightarrow$ Sum = 45.

$\Rightarrow$ Sum of averages coins in a box in a row or column = $45/3 = 15$ [The total number of coins in each row is the same. The total number of coins in each column is also the same.] $\Rightarrow$ (1)

Let us represent the final configuration of the sacks in boxes as follows:

	Table		
	C-1	C-2	C-3
R-1			
R-2			
R-3			

Consider bag (3,3)

0 sacks have more than 5 coins and ** => conditions i & ii are being satisfied.

=> 1,1,c are the coins. Now $c = 1$ or 2 or 3 or 4 => $c = 1$ or 4 for number of coins to be a multiple of 3.

But $c = 1$ as no other bag has the possibility to get avg. = 1 as bag (2,2) should have 1, b, c coins and b and c should be more than 1 as only 1*

=> bag (3,3) has 1, 1, 1 coins with average = 1.

Now, we can fill the averages in all the bags.

	Table		
	C-1	C-2	C-3
R-1	1,1,7 (3)	3,9,9 (7)	1,6,8 (5)
R-2	1,2,9 (4)	Avg = 2	Avg = 9
R-3	7,8,9 (8)	Avg = 6	1,1,1 (1)

In bag (2,3) Avg. = 9 => 9, 9, 9 are the coins.

In bag (2,2) => Avg. = 2 => Sum = 6 and only 1* => smallest elements should be 1.

=> 1, b, c are the coins where $b + c = 5$ and b,c can't be equal to 1 and less than 5 => $2 + 3 = 5$ is the only possibility.

=> 1, 2, 3 are the coins with average = 2.

	Table		
	C-1	C-2	C-3
R-1	1,1,7 (3)	3,9,9 (7)	1,6,8 (5)
R-2	1,2,9 (4)	1,2,3 (2)	9,9,9 (9)
R-3	7,8,9 (8)	Avg = 6	1,1,1 (1)

Considering bag (3,2)

Avg. = 6 => Sum = 18.

2 sacks more than 5 coins and ** => 2 sacks have 1 and 9 coins.

=> bag (3,2) has 1, c, 9 coins and $c = 18 - 1 - 9 = 8$

=> bag (3,2) has 1, 8, 9 coins with average = 6 coins.

=> Final required table, bracket number => average coins per sack in the bag.

	Table		
	C-1	C-2	C-3
R-1	1,1,7 (3)	3,9,9 (7)	1,6,8 (5)
R-2	1,2,9 (4)	1,2,3 (2)	9,9,9 (9)
R-3	7,8,9 (8)	1,8,9 (6)	1,1,1 (1)

Sum of coins in 3rd row = $8*3 + 6*3 + 1*3 = 45$.

 VIDEO SOLUTION

5.5

We are given that each box contains three sacks. Each sack has a certain number of coins, between 1 and 9, both inclusive.

The average number of coins per sack in the boxes are all distinct integers. The total number of coins in each row is the same. The total number of coins in each column is also the same.

=> The total number of coins in a box range from 3 ($1 + 1 + 1$) to 27 ($9 + 9 + 9$)

Since, it is given that the average number of coins per sack in the boxes are all distinct integers => The total number of coins in a box would be 3, 6, 9, 12, 15, 18, 21, 24, 27 => averages of 1, 2, 3, 4, ..., 9 => Sum = 45.

=> Sum of averages coins in a box in a row or column = $45/3 = 15$ [The total number of coins in each row is the same. The total number of coins in each column is also the same.] => (1)

Let us represent the final configuration of the sacks in boxes as follows:

	Table		
	C-1	C-2	C-3
R-1			
R-2			
R-3			

Also a bag (x,y) => bag in xth row and yth column.

We are given 2 clues => Table-1 & Table-2

Consider bag (3,1)

=> From Table-1 => Median = 8 & From Table-2 all 3 sacks have more than 5 coins. Also * => There is a 9 in one of the sacks.

=> c, 8, 9 are the coins in bag (3,1), now $c > 5$ & $c + 8 + 9$ should be a multiple of 3 => $c = 7$ is the only possibility.

=> bag (3,1) has 7, 8, 9 coins with average = 8.

Consider bag (2,1)

Median = 2 and 1 sack has more than 5 coins. Also ** => conditions i & iii should be satisfied.

=> 1, 2, 9 are the coins in bag (2,1) with average = 4

Consider bag (1,2)

Median = 9 and 2 elements are more than 5. Also * => (9 is present & 1 is not present)

=> c, 9, 9 are the coins in bag (1,2) and c is not equal to 1 and less than 5 => $c = 3$ for $c + 18$ to be a multiple of 3.

=> 3, 9, 9 are the coins in bag (1,2) with average = 7.

Capturing this info. in the table:

	Table		
	C-1	C-2	C-3
R-1		3,9,9 (7)	
R-2	1,2,9 (4)		
R-3	7,8,9 (8)		

From (1), The average in bag (1,1) is $15 - 4 - 8 = 3$.

From (1), The average in bag (1,3) is $15 - 3 - 7 = 5$.

	Table		
	C-1	C-2	C-3
R-1	Avg = 3	3,9,9 (7)	Avg = 5
R-2	1,2,9 (4)		
R-3	7,8,9 (8)		

Consider bag (1,1)

Avg = 3, 1 sack has more than 5 and ** => 2 conditions are being satisfied. => (can't be condition-3 => 9 coins as the total sum of coins is itself $3 \times 3 = 9$)

=> bag (1,1) has 1, 1, 7 coins with average = 3.

Consider bag (1,3)

Avg. = 5 => Sum = 15.

Median = 6 and 2 sacks have more than 5 and * => (1 condition is satisfied)

Not condition ii as the median is 6 & Not condition iii as the sum of 2 sacks itself will become $6 + 9 = 15$

=> 1, 6, c are the coins => For sum = 15 => $c = 15 - 1 - 6 = 8$

=> bag (1,3) has 1, 6, 8 coins with average = 5.

	Table		
	C-1	C-2	C-3
R-1	1,1,7 (3)	3,9,9 (7)	1,6,8 (5)
R-2	1,2,9 (4)		
R-3	7,8,9 (8)		

Consider bag (3,3)

0 sacks have more than 5 coins and ** => conditions i & ii are being satisfied.

=> 1,1,c are the coins. Now $c = 1$ or 2 or 3 or 4 => $c = 1$ or 4 for number of coins to be a multiple of 3.

But $c = 1$ as no other bag has the possibility to get avg. = 1 as bag (2,2) should have 1, b, c coins and b and c should be more than 1 as only 1*

=> bag (3,3) has 1, 1, 1 coins with average = 1.

Now, we can fill the averages in all the bags.

	Table		
	C-1	C-2	C-3
R-1	1,1,7 (3)	3,9,9 (7)	1,6,8 (5)
R-2	1,2,9 (4)	Avg = 2	Avg = 9
R-3	7,8,9 (8)	Avg = 6	1,1,1 (1)

In bag (2,3) Avg. = 9 => 9, 9, 9 are the coins.

In bag (2,2) => Avg. = 2 => Sum = 6 and only 1* => smallest element should be 1.

=> 1, b, c are the coins where $b + c = 5$ and b,c can't be equal to 1 and less than 5 => $2 + 3 = 5$ is the only possibility.

=> 1, 2, 3 are the coins with average = 2.

	Table		
	C-1	C-2	C-3
R-1	1,1,7 (3)	3,9,9 (7)	1,6,8 (5)
R-2	1,2,9 (4)	1,2,3 (2)	9,9,9 (9)
R-3	7,8,9 (8)	Avg = 6	1,1,1 (1)

Considering bag (3,2)

Avg. = 6 => Sum = 18.

2 sacks more than 5 coins and ** => 2 sacks have 1 and 9 coins.

=> bag (3,2) has 1, c, 9 coins and $c = 18 - 1 - 9 = 8$

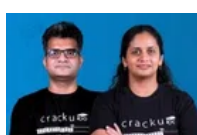
=> bag (3,2) has 1, 8, 9 coins with average = 6 coins.

=> Final required table, bracket number => average coins per sack in the bag.

	Table		
	C-1	C-2	C-3
R-1	1,1,7 (3)	3,9,9 (7)	1,6,8 (5)
R-2	1,2,9 (4)	1,2,3 (2)	9,9,9 (9)
R-3	7,8,9 (8)	1,8,9 (6)	1,1,1 (1)

Bags with different number of coins in all 3 sacks are (2,1), (3,2), (2,2), (3,2), (1,3) => 5 bags.

VIDEO SOLUTION



CAT Study Material

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Assuming the tickets from A - C, A - D, and A - E to be a , b and c , respectively. Also, assuming the tickets from C - D and C - E to be x and y , respectively. We are also given that the number of tickets from A - C is equal to the number of tickets booked from A - E, so the value of $a = c$. We are also given that 40 tickets were booked from B to C, and 30 tickets were booked from B to E. Other information provided is that no tickets were booked from A to B, from B to D, or from D to E.

Putting all the given information in the table, we get,

	A	B	C	D	E
A	0	0	a	b	$c = a$
B	0	0	40	0	30
C	0	0	0	x	y
D	0	0	0	0	0
E	0	0	0	0	0

From the above table, the tickets in the segments A - B, B - C, C - D and D - E can be calculated as,

$$A - B = a + b + c = 2a + b$$

$$B - C = a + b + c + 30 + 40 = 2a + b + 70$$

$$C - D = a + b + x + y + 30$$

$$D - E = c + y + 30 = a + y + 30$$

We are given that segment C - D had an occupancy factor of 95%. We can calculate the number of seats occupied in the segment C - D as,

$$\text{Seats occupied in segment C - D} = \frac{95}{100} \times 200 = 190$$

We are given that the B - C segment had more occupancy than the C - D segment. We are also given that a , b , c , x and y are multiples of 10 as per clue 6.

We also know that the people on board in a segment cannot be more than 200. The only value greater than 190 that is a multiple of 10 is 200. So, B - C segment had 200 seats occupied.

Equating it with the above equation, we get,

$$2a + b + 70 = 200$$

$$2a + b = 130 \text{ --(1)}$$

$$a + b + x + y + 30 = 190$$

$$a + b + x + y = 160 \text{ --(2)}$$

We are also given that among the seats reserved on segment D - E, exactly four-sevenths were from stations before C. The seats reserved on segment D - E before station C are $a + 30$.

Equating the expressions, we get,

$$\frac{4}{7}(a + y + 30) = a + 30$$

$$a + y + 30 = \frac{7}{4}(a + 30) \text{---(3)}$$

$$\text{So, the seats occupied during D - E} = a + y + 30 = \frac{7}{4}(a + 30)$$

We know that this value has to be an integer.

We are given that the number of tickets booked from A to C was higher than that from B to E. This means that the value of $a > 30$.

We know that a and b are positive multiples of 10, so the possible values of a that are greater than 30 and satisfy the equation (1) are 40, 50 and 60.

The only value of a at which the equation (3) is an integer is when $a + 30$ is a multiple of 4, and the only value out of the above values that satisfies the condition is when $a = 50$, as at $a = 40$ and 60, the value of $a + 30$ is not a multiple of 4.

We can conclude that the value of $a = 50$, and substituting in (1), we get the value of $b = 30$.

Substituting in (3), we get,

$$50 + y + 30 = \frac{7}{4}(50 + 30)$$

$$y + 80 = 140$$

$$y = 60$$

Substituting all the values in (2), we get,

$$50 + 30 + x + 60 = 160$$

$$x = 20$$

Putting all the values in the table, we get,

	A	B	C	D	E
A	0	0	50	30	50
B	0	0	40	0	30
C	0	0	0	20	60
D	0	0	0	0	0
E	0	0	0	0	0

$$A - B = a + b + c = 2a + b = 130$$

$$B - C = a + b + c + 30 + 40 = 2a + b + 70 = 200$$

$$C - D = a + b + x + y + 30 = 190$$

$$D - E = c + y + 30 = a + y + 30 = 140$$

The number of tickets booked from station A to E from the above table is 50.

Hence, the correct answer is 50.

[VIEW SOLUTION](#)

7. B

Assuming the tickets from A - C, A - D, and A - E to be a , b and c , respectively. Also, assuming the tickets from C - D and C - E to be x and y , respectively. We are also given that the number of tickets from A - C is equal to the number of tickets booked from A - E, so the value of $a = c$. We are also given that 40 tickets were booked from B to C, and 30 tickets were booked from B to E. Other information provided is that no tickets were booked from A to B, from B to D, or from D to E.

Putting all the given information in the table, we get,

	A	B	C	D	E
A	0	0	a	b	$c = a$
B	0	0	40	0	30
C	0	0	0	x	y
D	0	0	0	0	0
E	0	0	0	0	0

From the above table, the tickets in the segments A - B, B - C, C - D and D - E can be calculated as,

$$A - B = a + b + c = 2a + b$$

$$B - C = a + b + c + 30 + 40 = 2a + b + 70$$

$$C - D = a + b + x + y + 30$$

$$D - E = c + y + 30 = a + y + 30$$

We are given that segment C - D had an occupancy factor of 95%. We can calculate the number of seats occupied in the segment C - D as,

$$\text{Seats occupied in segment C - D} = \frac{95}{100} \times 200 = 190$$

We are given that the B - C segment had more occupancy than the C - D segment. We are also given that a , b , c , x and y are multiples of 10 as per clue 6.

We also know that the people on board in a segment cannot be more than 200. The only value greater than 190 that is a multiple of 10 is 200. So, B - C segment had 200 seats occupied.

Equating it with the above equation, we get,

$$2a + b + 70 = 200$$

$$2a + b = 130 \text{ ---(1)}$$

$$a + b + x + y + 30 = 190$$

$$a + b + x + y = 160 \text{ ---(2)}$$

We are also given that among the seats reserved on segment D - E, exactly four-sevenths were from stations before C. The seats reserved on segment D - E before station C are $a + 30$.

Equating the expressions, we get,

$$\frac{4}{7}(a + y + 30) = a + 30$$

$$a + y + 30 = \frac{7}{4}(a + 30) \quad \text{---(3)}$$

$$\text{So, the seats occupied during D - E} = a + y + 30 = \frac{7}{4}(a + 30)$$

We know that this value has to be an integer.

We are given that the number of tickets booked from A to C was higher than that from B to E. This means that the value of $a > 30$.

We know that a and b are positive multiples of 10, so the possible values of a that are greater than 30 and satisfy the equation (1) are 40, 50 and 60.

The only value of a at which the equation (3) is an integer is when $a + 30$ is a multiple of 4, and the only value out of the above values that satisfies the condition is when $a = 50$, as at $a = 40$ and 60, the value of $a + 30$ is not a multiple of 4.

We can conclude that the value of $a = 50$, and substituting in (1), we get the value of $b = 30$.

Substituting in (3), we get,

$$50 + y + 30 = \frac{7}{4}(50 + 30)$$

$$y + 80 = 140$$

$$y = 60$$

Substituting all the values in (2), we get,

$$50 + 30 + x + 60 = 160$$

$$x = 20$$

Putting all the values in the table, we get,

	A	B	C	D	E
A	0	0	50	30	50
B	0	0	40	0	30
C	0	0	0	20	60
D	0	0	0	0	0
E	0	0	0	0	0

$$A - B = a + b + c = 2a + b = 130$$

$$B - C = a + b + c + 30 + 40 = 2a + b + 70 = 200$$

$$C - D = a + b + x + y + 30 = 190$$

$$D - E = c + y + 30 = a + y + 30 = 140$$

Occupancy factor of segment D - E can be calculated as,

$$\text{Occupancy factor} = \frac{140}{200} \times 100 = 70\%$$

Hence, the correct answer is option B.

[VIEW SOLUTION](#)

8.80

Assuming the tickets from A - C, A - D, and A - E to be a , b and c , respectively. Also, assuming the tickets from C - D and C - E to be x and y , respectively. We are also given that the number of tickets from A - C is equal to the number of tickets booked from A - E, so the value of $a = c$. We are also given that 40 tickets were booked from B to C, and 30 tickets were booked from B to E. Other information provided is that no tickets were booked from A to B, from B to D, or from D to E.

Putting all the given information in the table, we get,

	A	B	C	D	E
A	0	0	a	b	c = a
B	0	0	40	0	30
C	0	0	0	x	y
D	0	0	0	0	0
E	0	0	0	0	0

From the above table, the tickets in the segments A - B, B - C, C - D and D - E can be calculated as,

$$A - B = a + b + c = 2a + b$$

$$B - C = a + b + c + 30 + 40 = 2a + b + 70$$

$$C - D = a + b + x + y + 30$$

$$D - E = c + y + 30 = a + y + 30$$

We are given that segment C - D had an occupancy factor of 95%. We can calculate the number of seats occupied in the segment C - D as,

$$\text{Seats occupied in segment C - D} = \frac{95}{100} \times 200 = 190$$

We are given that the B - C segment had more occupancy than the C - D segment. We are also given that a , b , c , x and y are multiples of 10 as per clue 6.

We also know that the people on board in a segment cannot be more than 200. The only value greater than 190 that is a multiple of 10 is 200. So, B - C segment had 200 seats occupied.

Equating it with the above equation, we get,

$$2a + b + 70 = 200$$

$$2a + b = 130 \text{ --(1)}$$

$$a + b + x + y + 30 = 190$$

$$a + b + x + y = 160 \text{ --(2)}$$

We are also given that among the seats reserved on segment D - E, exactly four-sevenths were from stations before C. The seats reserved on segment D - E before station C are $a + 30$.

Equating the expressions, we get,

$$\frac{4}{7}(a + y + 30) = a + 30$$

$$a + y + 30 = \frac{7}{4}(a + 30) \text{ --(3)}$$

$$\text{So, the seats occupied during D - E} = a + y + 30 = \frac{7}{4}(a + 30)$$

We know that this value has to be an integer.

We are given that the number of tickets booked from A to C was higher than that from B to E. This means that the value of $a > 30$.

We know that a and b are positive multiples of 10, so the possible values of a that are greater than 30 and satisfy the equation (1) are 40, 50 and 60.

The only value of a at which the equation (3) is an integer is when $a + 30$ is a multiple of 4, and the only value out of the above values that satisfies the condition is when $a = 50$, as at $a = 40$ and 60, the value of $a + 30$ is not a multiple of 4.

We can conclude that the value of $a = 50$, and substituting in (1), we get the value of $b = 30$.

Substituting in (3), we get,

$$50 + y + 30 = \frac{7}{4}(50 + 30)$$

$$y + 80 = 140$$

$$y = 60$$

Substituting all the values in (2), we get,

$$50 + 30 + x + 60 = 160$$

$$x = 20$$

Putting all the values in the table, we get,

	A	B	C	D	E
A	0	0	50	30	50
B	0	0	40	0	30
C	0	0	0	20	60
D	0	0	0	0	0
E	0	0	0	0	0

$$A - B = a + b + c = 2a + b = 130$$

$$B - C = a + b + c + 30 + 40 = 2a + b + 70 = 200$$

$$C - D = a + b + x + y + 30 = 190$$

$$D - E = c + y + 30 = a + y + 30 = 140$$

Number of tickets booked from station C = $20 + 60 = 80$.

Hence, the correct answer is 80.

[VIEW SOLUTION](#)

9.40

Assuming the tickets from A - C, A - D, and A - E to be a , b and c , respectively. Also, assuming the tickets from C - D and C - E to be x and y , respectively. We are also given that the number of tickets from A - C is equal to the number of tickets booked from A - E, so the value of $a = c$. We are also given that 40 tickets were booked from B to C, and 30 tickets were booked from B to E. Other information provided is that no tickets were booked from A to B, from B to D, or from D to E.

Putting all the given information in the table, we get,

	A	B	C	D	E
A	0	0	a	b	c = a
B	0	0	40	0	30
C	0	0	0	x	y
D	0	0	0	0	0
E	0	0	0	0	0

From the above table, the tickets in the segments A - B, B - C, C - D and D - E can be calculated as,

$$A - B = a + b + c = 2a + b$$

$$B - C = a + b + c + 30 + 40 = 2a + b + 70$$

$$C - D = a + b + x + y + 30$$

$$D - E = c + y + 30 = a + y + 30$$

We are given that segment C - D had an occupancy factor of 95%. We can calculate the number of seats occupied in the segment C - D as,

$$\text{Seats occupied in segment C - D} = \frac{95}{100} \times 200 = 190$$

We are given that the B - C segment had more occupancy than the C - D segment. We are also given that a, b, c, x and y are multiples of 10 as per clue 6.

We also know that the people on board in a segment cannot be more than 200. The only value greater than 190 that is a multiple of 10 is 200. So, B - C segment had 200 seats occupied.

Equating it with the above equation, we get,

$$2a + b + 70 = 200$$

$$2a + b = 130 \text{ --(1)}$$

$$a + b + x + y + 30 = 190$$

$$a + b + x + y = 160 \text{ --(2)}$$

We are also given that among the seats reserved on segment D - E, exactly four-sevenths were from stations before C. The seats reserved on segment D - E before station C are a + 30.

Equating the expressions, we get,

$$\frac{4}{7}(a + y + 30) = a + 30$$

$$a + y + 30 = \frac{7}{4}(a + 30) \text{ --(3)}$$

$$\text{So, the seats occupied during D - E} = a + y + 30 = \frac{7}{4}(a + 30)$$

We know that this value has to be an integer.

We are given that the number of tickets booked from A to C was higher than that from B to E. This means that the value of $a > 30$.

We know that a and b are positive multiples of 10, so the possible values of a that are greater than 30 and satisfy the equation (1) are 40, 50 and 60.

The only value of a at which the equation (3) is an integer is when $a + 30$ is a multiple of 4, and the only value out of the above values that satisfies the condition is when $a = 50$, as at $a = 40$ and 60, the value of $a + 30$ is not a multiple of 4.

We can conclude that the value of $a = 50$, and substituting in (1), we get the value of $b = 30$.

Substituting in (3), we get,

$$50 + y + 30 = \frac{7}{4}(50 + 30)$$

$$y + 80 = 140$$

$$y = 60$$

Substituting all the values in (2), we get,

$$50 + 30 + x + 60 = 160$$

$$x = 20$$

Putting all the values in the table, we get,

	A	B	C	D	E
A	0	0	50	30	50
B	0	0	40	0	30
C	0	0	0	20	60
D	0	0	0	0	0
E	0	0	0	0	0

$$A - B = a + b + c = 2a + b = 130$$

$$B - C = a + b + c + 30 + 40 = 2a + b + 70 = 200$$

$$C - D = a + b + x + y + 30 = 190$$

$$D - E = c + y + 30 = a + y + 30 = 140$$

The number of tickets booked to station C = $50 + 40 = 90$

The number of tickets booked to station D = $30 + 20 = 50$

Difference = $90 - 50 = 40$.

Hence, the correct answer is 40.

[VIEW SOLUTION](#)

10.60

Assuming the tickets from A - C, A - D, and A - E to be a , b and c , respectively. Also, assuming the tickets from C - D and C - E to be x and y , respectively. We are also given that the number of tickets from A - C is equal to the number of tickets booked from A - E, so the value of $a = c$. We are also given that 40 tickets were booked from B to C, and 30 tickets were booked from B to E. Other information provided is that no tickets were booked from A to B, from B to D, or from D to E.

Putting all the given information in the table, we get,

	A	B	C	D	E
A	0	0	a	b	$c = a$
B	0	0	40	0	30
C	0	0	0	x	y
D	0	0	0	0	0
E	0	0	0	0	0

From the above table, the tickets in the segments A - B, B - C, C - D and D - E can be calculated as,

$$A - B = a + b + c = 2a + b$$

$$B - C = a + b + c + 30 + 40 = 2a + b + 70$$

$$C - D = a + b + x + y + 30$$

$$D - E = c + y + 30 = a + y + 30$$

We are given that segment C - D had an occupancy factor of 95%. We can calculate the number of seats occupied in the segment C - D as,

$$\text{Seats occupied in segment C - D} = \frac{95}{100} \times 200 = 190$$

We are given that the B - C segment had more occupancy than the C - D segment. We are also given that a , b , c , x and y are multiples of 10 as per clue 6.

We also know that the people on board in a segment cannot be more than 200. The only value greater than 190 that is a multiple of 10 is 200. So, B - C segment had 200 seats occupied.

Equating it with the above equation, we get,

$$2a + b + 70 = 200$$

$$2a + b = 130 \text{ --(1)}$$

$$a + b + x + y + 30 = 190$$

$$a + b + x + y = 160 \text{ --(2)}$$

We are also given that among the seats reserved on segment D - E, exactly four-sevenths were from stations before C. The seats reserved on segment D - E before station C are $a + 30$.

Equating the expressions, we get,

$$\frac{4}{7}(a + y + 30) = a + 30$$

$$a + y + 30 = \frac{7}{4}(a + 30) \text{---(3)}$$

$$\text{So, the seats occupied during D - E} = a + y + 30 = \frac{7}{4}(a + 30)$$

We know that this value has to be an integer.

We are given that the number of tickets booked from A to C was higher than that from B to E. This means that the value of $a > 30$.

We know that a and b are positive multiples of 10, so the possible values of a that are greater than 30 and satisfy the equation (1) are 40, 50 and 60.

The only value of a at which the equation (3) is an integer is when $a + 30$ is a multiple of 4, and the only value out of the above values that satisfies the condition is when $a = 50$, as at $a = 40$ and 60, the value of $a + 30$ is not a multiple of 4.

We can conclude that the value of $a = 50$, and substituting in (1), we get the value of $b = 30$.

Substituting in (3), we get,

$$50 + y + 30 = \frac{7}{4}(50 + 30)$$

$$y + 80 = 140$$

$$y = 60$$

Substituting all the values in (2), we get,

$$50 + 30 + x + 60 = 160$$

$$x = 20$$

Putting all the values in the table, we get,

	A	B	C	D	E
A	0	0	50	30	50
B	0	0	40	0	30
C	0	0	0	20	60
D	0	0	0	0	0
E	0	0	0	0	0

$$A - B = a + b + c = 2a + b = 130$$

$$B - C = a + b + c + 30 + 40 = 2a + b + 70 = 200$$

$$C - D = a + b + x + y + 30 = 190$$

$$D - E = c + y + 30 = a + y + 30 = 140$$

The number of tickets booked for exactly one segment can be calculated using values from the above table as,

Tickets booked for exactly one segment = $A - B + B - C + C - D + D - E = 0 + 40 + 20 + 0 = 60$.

Hence, the correct answer is 60.

[VIEW SOLUTION](#)



11. B

Number of flowers after the first dipping = $2x$

Number of flowers after the second dipping = $2(2x-y) = 4x-2y$

Number of flowers after the third dipping = $2(4x-2y-y) = 8x-6y$

Number of flowers after the fourth dipping = $2(8x-6y-y) = 16x-14y$

$16x-14y = y$

$y = 16x/15$

Minimum value of $y = 16$ when $x = 15$

[VIDEO SOLUTION](#)

12. C

Number of flowers after the first dipping = $2x$

Number of flowers after the second dipping = $2(2x-y) = 4x-2y$

Number of flowers after the third dipping = $2(4x-2y-y) = 8x-6y$

Number of flowers after the fourth dipping = $2(8x-6y-y) = 16x-14y$

$16x-14y = y$

$y = 16x/15$

Minimum value of $y = 16$ when $x = 15$

[VIDEO SOLUTION](#)

13. C

Number of flowers after the first dipping = 60

Number of flowers after the second dipping = $2(60-y) = 120-2y$

Number of flowers after the third dipping = $2(120-2y-y) = 240-6y$

Number of flowers after the fourth dipping = $2(240-6y-y) = 480-14y$

$y = 480 - 14y$

$15y = 480$

$y = 32$

[VIDEO SOLUTION](#)

14. 4

If all the two-author-type books are counted for both authors separately, then we get the sum to be $4 * 2 = 8$.

If all the three-author-type books are counted for all three authors separately, then we get the sum to be $3 * 3 = 9$.

We are given that there are 2 four-author books, so there are exactly two books for each that are of the four-author type.

Author	Single-author	Two-author	Three-author	Four-author	Total
Arman				2	5
Brajen				2	8
Chintan				2	12
Devon				2	10
	10	8	9	8	

Author	Single-author	Two-author	Three-author	Four-author	Total
Arman	1	1	1	2	5
Brajen	2	2	2	2	8
Chintan			3	2	12
Devon			3	2	10
	10	8	9	8	

Placing both possibilities in the table, we get,

Author	Single-author	Two-author	Three-author	Four-author	Total
Arman	1	1	1	2	5
Brajen	2	2	2	2	8
Chintan	4/3	3/4	3	2	12
Devon	3/4	2/1	3	2	10
	10	8	9	8	

The total number of two-author and three-author papers written by Brajen = $2 + 2 = 4$.

Hence, the correct answer is 4.

[VIEW SOLUTION](#)

15.A

If all the two-author-type books are counted for both authors separately, then we get the sum to be $4 * 2 = 8$.

If all the three-author-type books are counted for all three authors separately, then we get the sum to be $3 * 3 = 9$.

If all the four-author-type books are counted for all four authors separately, then we get the sum to be $4 * 2 = 8$.

We are given that there are 2 four-author books, so there are exactly two books for each that are of the four-author type.

Putting the values in the table, we get,

Author	Single-author	Two-author	Three-author	Four-author	Total
Arman				2	5
Brajen				2	8
Chintan				2	12
Devon				2	10
	10	8	9	8	

We are given that each author has at least one book of each type. In the case of Arman, there are three books left to be assigned, and there are 3 types of books left, so each of them has to be equal to 1 for the above condition to be satisfied.

In the case of Brajen, we are told that he had an equal number of single-author and two-author books.

Case 1: If the equal number is 1, then the three-author type books become 4, and we are given that both Chintan and Devon had more three-author books than Brajen, which is not possible in this case, as both of them combined will be left with 5 books together and whatever we split 5 books there won't be a case of both of them having more books than Brajen.

Case 2: If the equal number is 2, then the three-author type books become 2, and we are given that both Chintan and Devon had more three-author books than Brajen, which is possible in this case, as both of them combined will be left with 6 books together and the only way to split 6 books such that both of them are greater than 2 is to split them into 3, 3 and it is the only possible way.

Case 3: If the equal number is greater than or equal to 3, then the three-author type books become 0 or negative, which can directly be eliminated, as we are given that all authors wrote at least 1 book of each type.

So, we can eliminate both cases 1 and 3, and we will be left with the case of Brajen having 2 books each of Single-author, Two-author, and Three-author types. Chintan and Devon have three books each of the three-author type.

Filling up the table with these values, we get,

Author	Single-author	Two-author	Three-author	Four-author	Total
Arman	1	1	1	2	5
Brajen	2	2	2	2	8
Chintan			3	2	12
Devon			3	2	10
	10	8	9	8	

We are given that each author has a different number of single-author type books. The only possibilities for the single-author books of Chintan and Devon are 3 and 4 in some order.

If the single-author books of Chintan are 3, then the double-author books of Chintan become 4, the single-author books of Devon become 4, and the double-author books of Devon become 1.

If the single-author books of Chintan are 4, then the double-author books of Chintan become 3, the single-author books of Devon become 3, and the double-author books of Devon become 2.

Placing both possibilities in the table, we get,

Author	Single-author	Two-author	Three-author	Four-author	Total
Arman	1	1	1	2	5
Brajen	2	2	2	2	8
Chintan	4/3	3/4	3	2	12
Devon	3/4	2/1	3	2	10
	10	8	9	8	

Statement 1: Chintan wrote exactly three two-author papers.

This is not necessarily true, as he could have written four books as well.

Statement 2: Chintan wrote more single-author papers than Devon.

This is not necessarily true, as the opposite can be possible.

So, neither 1 nor 2 is necessarily true.

Hence, the correct answer is option A.

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16. B

If all the two-author-type books are counted for both authors separately, then we get the sum to be $4 * 2 = 8$.

If all the three-author-type books are counted for all three authors separately, then we get the sum to be $3 * 3 = 9$.

If all the four-author-type books are counted for all four authors separately, then we get the sum to be $4 * 2 = 8$.

We are given that there are 2 four-author books, so there are exactly two books for each that are of the four-author type.

Putting the values in the table, we get,

Author	Single-author	Two-author	Three-author	Four-author	Total
Arman				2	5
Brajen				2	8
Chintan				2	12
Devon				2	10
	10	8	9	8	

We are given that each author has at least one book of each type. In the case of Arman, there are three books left to be assigned, and there are 3 types of books left, so each of them has to be equal to 1 for the above condition to be satisfied.

In the case of Brajen, we are told that he had an equal number of single-author and two-author books.

Case 1: If the equal number is 1, then the three-author type books become 4, and we are given that both Chintan and Devon had more three-author books than Brajen, which is not possible in this case, as both of them combined will be left with 5 books together and whatever we split 5 books there won't be a case of both of them having more books than Brajen.

Case 2: If the equal number is 2, then the three-author type books become 2, and we are given that both Chintan and Devon had more three-author books than Brajen, which is possible in this case, as both of them combined will be left with 6 books together and the only way to split 6 books such that both of them are greater than 2 is to split them into 3, 3 and it is the only possible way.

Case 3: If the equal number is greater than or equal to 3, then the three-author type books become 0 or negative, which can directly be eliminated, as we are given that all authors wrote at least 1 book of each type.

So, we can eliminate both cases 1 and 3, and we will be left with the case of Brajen having 2 books each of Single-author, Two-author, and Three-author types. Chintan and Devon have three books each of the three-author type.

Filling up the table with these values, we get,

Author	Single-author	Two-author	Three-author	Four-author	Total
Arman	1	1	1	2	5
Brajen	2	2	2	2	8
Chintan			3	2	12
Devon			3	2	10
	10	8	9	8	

We are given that each author has a different number of single-author type books. The only possibilities for the single-author books of Chintan and Devon are 3 and 4 in some order.

If the single-author books of Chintan are 3, then the double-author books of Chintan become 4, the single-author books of Devon become 4, and the double-author books of Devon become 1.

If the single-author books of Chintan are 4, then the double-author books of Chintan become 3, the single-author books of Devon become 3, and the double-author books of Devon become 2.

Placing both possibilities in the table, we get,

Author	Single-author	Two-author	Three-author	Four-author	Total
Arman	1	1	1	2	5
Brajen	2	2	2	2	8
Chintan	4/3	3/4	3	2	12
Devon	3/4	2/1	3	2	10
	10	8	9	8	

Statement 1: Arman wrote three-author papers only with Chintan and Devon.

There are in total 3 three-author books, and Chintan and Devon are authors of all three of them according to the table. So, Arman wrote the three-author papers only with Chintan and Devon. So, the statement is necessarily true

Statement 2: Brajen wrote three-author papers only with Chintan and Devon.

There are in total 3 three-author books, and Chintan and Devon are authors of all three of them according to the table. So, Brajen wrote the three-author papers only with Chintan and Devon. So, the statement is necessarily true.

So, both 1 and 2 are necessarily true.

Hence, the correct answer is option B.

[VIEW SOLUTION](#)

17.3

If all the two-author-type books are counted for both authors separately, then we get the sum to be $4 * 2 = 8$.

If all the three-author-type books are counted for all three authors separately, then we get the sum to be $3 * 3 = 9$.

If all the four-author-type books are counted for all four authors separately, then we get the sum to be $4 * 2 = 8$.

We are given that there are 2 four-author books, so there are exactly two books for each that are of the four-author type.

Putting the values in the table, we get,

Author	Single-author	Two-author	Three-author	Four-author	Total
Arman				2	5
Brajen				2	8
Chintan				2	12
Devon				2	10
	10	8	9	8	

We are given that each author has at least one book of each type. In the case of Arman, there are three books left to be assigned, and there are 3 types of books left, so each of them has to be equal to 1 for the above condition to be satisfied.

In the case of Brajen, we are told that he had an equal number of single-author and two-author books.

Case 1: If the equal number is 1, then the three-author type books become 4, and we are given that both Chintan and Devon had more three-author books than Brajen, which is not possible in this case, as both of them combined will be left with 5 books together and whatever we split 5 books there wont be a case of both of them having more books than Brajen.

Case 2: If the equal number is 2, then the three-author type books become 2, and we are given that both Chintan and Devon had more three-author books than Brajen, which is possible in this case, as both of them combined will be left with 6 books together and the only way to split 6 books such that both of them are greater than 2 is to split them into 3, 3 and it is the only possible way.

Case 3: If the equal number is greater than or equal to 3, then the three-author type books become 0 or negative, which can directly be eliminated, as we are given that all authors wrote at least 1 book of each type.

So, we can eliminate both cases 1 and 3, and we will be left with the case of Brajen having 2 books each of Single-author, Two-author, and Three-author types. Chintan and Devon have three books each of the three-author type.

Filling up the table with these values, we get,

Author	Single-author	Two-author	Three-author	Four-author	Total
Arman	1	1	1	2	5
Brajen	2	2	2	2	8
Chintan			3	2	12
Devon			3	2	10
	10	8	9	8	

We are given that each author has a different number of single-author type books. The only possibilities for the single-author books of Chintan and Devon are 3 and 4 in some order.

If the single-author books of Chintan are 3, then the double-author books of Chintan become 4, the single-author books of Devon become 4, and the double-author books of Devon become 1.

If the single-author books of Chintan are 4, then the double-author books of Chintan become 3, the single-author books of Devon become 3, and the double-author books of Devon become 2.

Placing both possibilities in the table, we get,

Author	Single-author	Two-author	Three-author	Four-author	Total
Arman	1	1	1	2	5
Brajen	2	2	2	2	8
Chintan	4/3	3/4	3	2	12
Devon	3/4	2/1	3	2	10
	10	8	9	8	

If Devon wrote more than one two-author paper, he would have written 2 papers, making the number of two-author papers written by Chintan 3.

Hence, the correct answer is 3.

[VIEW SOLUTION](#)

18. D

Let the number of students in non-CS be $7x$ so no. of students taking AI in non-CS is $2x$ and no. of students taking ML in non-CS is $5x$

All the CS students have taken both the courses, we are given this in the first line of the set.

From statement 5, we can conclude that 0 students from CS failed in AI and 0 students from non-CS got grade A in AI.

From statement 6, let us say that the numbers of CS students who got A, B and C grades respectively in AI were in the ratio $3a$, $5a$ and $2a$, while in ML the ratio was $4b$, $5b$ and $2b$.

From statement 3 we can say that the number of non-CS students who failed in AI and ML are b in each category.

Now from statement 8, we can say that number of CS students who failed in ML is equal to $30-b$.

		A	B	C	F	Total
CS	AI	3a	5a	2a	0	10a
	ML	4b	5b	2b	30-b	10b+30
Non CS	AI	0			b	2x
	ML				b	5x

From statement 7, we can say that, $\frac{2b}{30-b} = \frac{3}{1}$

or, $2b = 90-3b$

or, $b = 18$

CS students take both the AI and ML courses, therefore $10a = 10b + 30$

or, $a = b+3 = 21$

From statement 2, we can say that $10a = 7x$

or, $x = 30$

Substituting the values of a, b and x in the above table.

		A	B	C	F	Total
CS	AI	63	105	42	0	210
	ML	72	90	36	12	210
Non CS	AI	0			18	60
	ML				18	150

A total of 270 students took AI out of which 252 students passed and a total of 360 students took ML out of which 330 students passed.

From statement 4, we can say that out of the 252 students who passed in AI, 126 of them got Grade B and 63 got Grade A and 63 got Grade C.

Similarly, We can say that out of the 330 students who passed in ML, 165 of them got Grade C and 99 got Grade A and 66 got Grade C.

Therefore, the final table which we get is

		A	B	C	F	Total
CS	AI	63	105	42	0	210
	ML	72	90	36	12	210
Non CS	AI	0	21	21	18	60
	ML	27	75	30	18	150

VIDEO SOLUTION

19.12

Let the number of students in non-CS be $7x$ so no. of students taking AI in non-CS is $2x$ and no. of students taking ML in non-CS is $5x$

All the CS students have taken both the courses, we are given this in the first line of the set.

From statement 5, we can conclude that 0 students from CS failed in AI and 0 students from non-CS got grade A in AI.

From statement 6, let us say that the numbers of CS students who got A, B and C grades respectively in AI were in the ratio $3a$, $5a$ and $2a$, while in ML the ratio was $4b$, $5b$ and $2b$.

From statement 3 we can say that the number of non-CS students who failed in AI and ML are b in each category.

Now from statement 8, we can say that number of CS students who failed in ML is equal to $30-b$.

		A	B	C	F	Total
CS	AI	$3a$	$5a$	$2a$	0	$10a$
	ML	$4b$	$5b$	$2b$	$30-b$	$10b+30$
Non CS	AI	0			b	$2x$
	ML				b	$5x$

From statement 7, we can say that, $\frac{2b}{30-b} = \frac{3}{1}$

or, $2b = 90-3b$

or, $b = 18$

CS students take both the AI and ML courses, therefore $10a = 10b + 30$

or, $a = b+3 = 21$

From statement 2, we can say that $10a = 7x$

or, $x = 30$

Substituting the values of a, b and x in the above table.

		A	B	C	F	Total
CS	AI	63	105	42	0	210
	ML	72	90	36	12	210
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Similarly, We can say that out of the 330 students who passed in ML, 165 of them got Grade C and 99 got Grade A and 66 got Grade C.

Therefore, the final table which we get is

		A	B	C	F	Total
CS	AI	63	105	42	0	210
	ML	72	90	36	12	210
Non CS	AI	0	21	21	18	60
	ML	27	75	30	18	150

VIDEO SOLUTION

20.27

Let the number of students in non-CS be $7x$ so no. of students taking AI in non-CS is $2x$ and no. of students taking ML in non-CS is $5x$

All the CS students have taken both the courses, we are given this in the first line of the set.

From statement 5, we can conclude that 0 students from CS failed in AI and 0 students from non-CS got grade A in AI.

From statement 6, let us say that the numbers of CS students who got A, B and C grades respectively in AI were in the ratio $3a$, $5a$ and $2a$, while in ML the ratio was $4b$, $5b$ and $2b$.

From statement 3 we can say that the number of non-CS students who failed in AI and ML are b in each category.

Now from statement 8, we can say that number of CS students who failed in MI is equal to 30-b.

		A	B	C	F	Total
CS	AI	3a	5a	2a	0	10a
	ML	4b	5b	2b	30-b	10b+30
Non CS	AI	0			b	2x
	ML				b	5x

From statement 7, we can say that, $\frac{2b}{30-b} = \frac{3}{1}$

or, $2b = 90-3b$

or, $b = 18$

CS students take both the AI and ML courses, therefore $10a = 10b + 30$

or, $a = b+3 = 21$

From statement 2, we can say that $10a = 7x$

or, $x = 30$

Substituting the values of a, b and x in the above table.

		A	B	C	F	Total
CS	AI	63	105	42	0	210
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A total of 270 students took AI out of which 252 students passed and a total of 360 students took ML out of which 330 students passed.

From statement 4, we can say that out of the 252 students who passed in AI, 126 of them got Grade B and 63 got Grade A and 63 got Grade C.

Similarly, We can say that out of the 330 students who passed in ML, 165 of them got Grade C and 99 got Grade A and 66 got Grade B.

Therefore, the final table which we get is

		A	B	C	F	Total
CS	AI	63	105	42	0	210
	ML	72	90	36	12	210
Non CS	AI	0	21	21	18	60
	ML	27	75	30	18	150

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21. D

Let the number of students in non-CS be $7x$ so no. of students taking AI in non-CS is $2x$ and no. of students taking ML in non-CS is $5x$

All the CS students have taken both the courses, we are given this in the first line of the set.

From statement 5, we can conclude that 0 students from CS failed in AI and 0 students from non-CS got grade A in AI.

From statement 6, let us say that the numbers of CS students who got A, B and C grades respectively in AI were in the ratio $3a$, $5a$ and $2a$, while in ML the ratio was $4b$, $5b$ and $2b$.

From statement 3 we can say that the number of non-CS students who failed in AI and ML are b in each category.

Now from statement 8, we can say that number of CS students who failed in ML is equal to $30-b$.

		A	B	C	F	Total
CS	AI	$3a$	$5a$	$2a$	0	$10a$
	ML	$4b$	$5b$	$2b$	$30-b$	$10b+30$
Non CS	AI	0			b	$2x$
	ML				b	$5x$

From statement 7, we can say that, $\frac{2b}{30-b} = \frac{3}{1}$

or, $2b = 90-3b$

or, $b = 18$

CS students take both the AI and ML courses, therefore $10a = 10b + 30$

or, $a = b + 3 = 21$

From statement 2, we can say that $10a = 7x$

or, $x = 30$

Substituting the values of a , b and x in the above table.

		A	B	C	F	Total
CS	AI	63	105	42	0	210
	ML	72	90	36	12	210
Non CS	AI	0			18	60
	ML				18	150

A total of 270 students took AI out of which 252 students passed and a total of 360 students took ML out of which 330 students passed.

From statement 4, we can say that out of the 252 students who passed in AI, 126 of them got Grade B and 63 got Grade A and 63 got Grade C.

Similarly, We can say that out of the 330 students who passed in ML, 165 of them got Grade C and 99 got Grade A and 66 got Grade B.

Therefore, the final table which we get is

		A	B	C	F	Total
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	ML	72	90	36	12	210
Non CS	AI	0	21	21	18	60
	ML	27	75	30	18	150

▶ VIDEO SOLUTION

22. B

Let the number of students in non-CS be $7x$ so no. of students taking AI in non-CS is $2x$ and no. of students taking ML in non-CS is $5x$

All the CS students have taken both the courses, we are given this in the first line of the set.

From statement 5, we can conclude that 0 students from CS failed in AI and 0 students from non-CS got grade A in AI.

From statement 6, let us say that the numbers of CS students who got A, B and C grades respectively in AI were in the ratio 3a, 5a and 2a, while in ML the ratio was 4b, 5b and 2b.

From statement 3 we can say that the number of non- CS students who failed in AI and ML are b in each category.

Now from statement 8, we can say that number of CS students who failed in ML is equal to 30-b.

		A	B	C	F	Total
CS	AI	3a	5a	2a	0	10a
	ML	4b	5b	2b	30-b	10b+30
Non CS	AI	0			b	2x
	ML				b	5x

From statement 7, we can say that, $\frac{2b}{30-b} = \frac{3}{1}$

or, $2b = 90-3b$

or, $b = 18$

CS students take both the AI and ML courses, therefore $10a = 10b + 30$

or, $a = b+3 = 21$

From statement 2, we can say that $10a = 7x$

or, $x = 30$

Substituting the values of a, b and x in the above table.

		A	B	C	F	Total
CS	AI	63	105	42	0	210
	ML	72	90	36	12	210
Non CS	AI	0			18	60
	ML				18	150

A total of 270 students took AI out of which 252 students passed and a total of 360 students took ML out of which 330 students passed.

From statement 4, we can say that out of the 252 students who passed in AI, 126 of them got Grade B and 63 got Grade A and 63 got Grade C.

Similarly, We can say that out of the 330 students who passed in ML, 165 of them got Grade C and 99 got Grade A and 66 got Grade C.

Therefore, the final table which we get is

		A	B	C	F	Total
CS	AI	63	105	42	0	210
	ML	72	90	36	12	210
Non CS	AI	0	21	21	18	60
	ML	27	75	30	18	150

VIDEO SOLUTION

23.26

Table 1: 2-day averages for Days 2 through 5			
Day 2	Day 3	Day 4	Day 5
15	15.5	16	17

Let the total score of day 1, day 2, day 3, day 4, and day 5 are d_1 , d_2 , d_3 , d_4 , and d_5 , respectively.

The table shows that $d_1 + d_2 = 30 \dots \text{eq (1)}$, $d_2 + d_3 = 31 \dots \text{eq (2)}$, $d_3 + d_4 = 32 \dots \text{eq(3)}$, $d_4 + d_5 = 34 \dots \text{eq(4)}$

It is given that participants are ranked each day, with the person having the maximum score being awarded the minimum rank (1) on that day. All participants with a tied score are awarded the best available rank if there is a tie.

It is given that the total score on Day 3 is the same as the total score on Day 4.

Therefore, $d_3 = d_4 \Rightarrow d_3 = d_4 = 16$, which implies $d_2 = 15$, $d_5 = 18$, and $d_1 = 15$.

The day-wise score is given below:

	Day 1	Day 2	Day 3	Day 4	Day 5
Akhil					
Bimal					
Chatur					
Total Score	15	15	16	16	18

It is known that Chatur always scores in multiples of 3. His score on Day 2 is the unique highest score in the competition. His minimum score is observed only on Day 1, and it matches Akhil's score on Day 4.

Hence, only Chatur scored 9 (one time) on Day 2, and no other person scored 9 on any of the given 5 days. Chatur scored 3 only one time, which was on Day 1. Therefore, the scores obtained by Chatur on Day 3, Day 4, and Day 5 are 6, 6, and 6, respectively. It is also known that Akhil's score on Day 4 is the same as the score obtained by Chatur on Day 1. Hence, Akhil's score on Day 4 is 3.

Hence, we get the following table:

	Day 1	Day 2	Day 3	Day 4	Day 5
Akhil				3	
Bimal				7	
Chatur	3	9	6	6	6
Total Score	15	15	16	16	18

From Table 2, we see that the rank of Bimal and Akhil is the same, which is 2. Hence, The score obtained by Akhil and Bimal is the same. Let the score be x . Therefore, $6+2x = 16 \Rightarrow x = 5$

The rank of Chatur on Day 5 is 2, and the rank of Bimal is 1, which implies the score obtained by Bimal will be more than Chatur. Hence, Bimal can score either 7 or 8 on Day 5. Therefore, the score obtained by Akhil on Day 5 is either 5 or 4.

	Day 1	Day 2	Day 3	Day 4	Day 5
Akhil			5	3	5/4
Bimal			5	7	7/8
Chatur	3	9	6	6	6
Total Score	15	15	16	16	18

It is given that Bimal's scores are the same on Day 1 and Day 3. Hence, the score obtained by Bimal on Day 1 is 5, which implies The score obtained by Akhil is 7 on Day 1.

From Table 2, we can see that the rank of Bimal is 3 on Day 2, and the rank of Akhil is 2 on Day 2. Hence, the score of Bimal will be lower than Akhil on Day 2.

Let the score of Akhil be a , and the score of Bimal be b . Then $9+a+b = 15$, and $a > b$

$\Rightarrow a+b=6$, and $a > b$

Hence, the value of a can be $4/5$, and the value of b can be $2/1$

Therefore, the final table is given below:

	Day 1	Day 2	Day 3	Day 4	Day 5	Total score
Akhil	7	4/5	5	3	5/4	23/24/25
Bimal	5	2/1	5	7	7/8	27/26/25
Chatur	3	9	6	6	6	30
Total Score	15	15	16	16	18	80

In the question, it is given that the score obtained by Akhil is 24, which implies the score obtained by Bimal is 26.

The answer is 26

[VIDEO SOLUTION](#)

24.25

Table 1: 2-day averages for Days 2 through 5			
Day 2	Day 3	Day 4	Day 5
15	15.5	16	17

Let the total score of day 1, day 2, day 3, day 4, and day 5 are d_1 , d_2 , d_3 , d_4 , and d_5 , respectively.

The table shows that $d_1+d_2 = 30 \dots \text{eq (1)}$, $d_2+d_3 = 31 \dots \text{eq (2)}$, $d_3+d_4 = 32 \dots \text{eq(3)}$, $d_4+d_5 = 34 \dots \text{eq(4)}$

It is given that participants are ranked each day, with the person having the maximum score being awarded the minimum rank (1) on that day. All participants with a tied score are awarded the best available rank if there is a tie.

It is given that the total score on Day 3 is the same as the total score on Day 4.

Therefore, $d_3 = d_4 \Rightarrow d_3 = d_4 = 16$, which implies $d_2 = 15$, $d_5 = 18$, and $d_1 = 15$.

The day-wise score is given below:

	Day 1	Day 2	Day 3	Day 4	Day 5
Akhil					
Bimal					
Chatur					
Total Score	15	15	16	16	18

It is known that Chatur always scores in multiples of 3. His score on Day 2 is the unique highest score in the competition. His minimum score is observed only on Day 1, and it matches Akhil's score on Day 4.

Hence, only Chatur scored 9 (one time) on Day 2, and no other person scored 9 on any of the given 5 days. Chatur scored 3 only one time, which was on Day 1. Therefore, the scores obtained by Chatur on Day 3, Day 4, and Day 5 are 6, 6, and 6, respectively. It is also known that Akhil's score on Day 4 is the same as the score obtained by Chatur on Day 1. Hence, Akhil's score on Day 4 is 3.

Hence, we get the following table:

	Day 1	Day 2	Day 3	Day 4	Day 5
Akhil				3	
Bimal				7	
Chatur	3	9	6	6	6
Total Score	15	15	16	16	18

From Table 2, we see that the rank of Bimal and Akhil is the same, which is 2. Hence, The score obtained by Akhil and Bimal is the same. Let the score be x . Therefore, $6+2x = 16 \Rightarrow x = 5$

The rank of Chatur on Day 5 is 2, and the rank of Bimal is 1, which implies the score obtained by Bimal will be more than Chatur. Hence, Bimal can score either 7 or 8 on Day 5. Therefore, the score obtained by Akhil on Day 5 is either 5 or 4.

	Day 1	Day 2	Day 3	Day 4	Day 5
Akhil			5	3	5/4
Bimal			5	7	7/8
Chatur	3	9	6	6	6
Total Score	15	15	16	16	18

It is given that Bimal's scores are the same on Day 1 and Day 3. Hence, the score obtained by Bimal on Day 1 is 5, which implies The score obtained by Akhil is 7 on Day 1.

From Table 2, we can see that the rank of Bimal is 3 on Day 2, and the rank of Akhil is 2 on Day 2. Hence, the score of Bimal will be lower than Akhil on Day 2.

Let the score of Akhil be a , and the score of Bimal be b . Then $9+a+b = 15$, and $a > b$

$\Rightarrow a+b=6$, and $a > b$

Hence, the value of a can be $4/5$, and the value of b can be $2/1$

Therefore, the final table is given below:

	Day 1	Day 2	Day 3	Day 4	Day 5	Total score
Akhil	7	4/5	5	3	5/4	23/24/25
Bimal	5	2/1	5	7	7/8	27/26/25
Chatur	3	9	6	6	6	30
Total Score	15	15	16	16	18	80

From the table, we can see that the minimum score obtained by Bimal is 25.

VIDEO SOLUTION

25. D

Table 1: 2-day averages for Days 2 through 5			
Day 2	Day 3	Day 4	Day 5
15	15.5	16	17

Let the total score of day 1, day 2, day 3, day 4, and day 5 are d_1 , d_2 , d_3 , d_4 , and d_5 , respectively.

The table shows that $d_1 + d_2 = 30$... eq (1), $d_2 + d_3 = 31$... eq (2), $d_3 + d_4 = 32$ eq(3), $d_4 + d_5 = 34$... eq(4)

It is given that participants are ranked each day, with the person having the maximum score being awarded the minimum rank (1) on that day. All participants with a tied score are awarded the best available rank if there is a tie.

It is given that the total score on Day 3 is the same as the total score on Day 4.

Therefore, $d_3 = d_4 \Rightarrow d_3 = d_4 = 16$, which implies $d_2 = 15$, $d_5 = 18$, and $d_1 = 15$.

The day-wise score is given below:

	Day 1	Day 2	Day 3	Day 4	Day 5
Akhil					
Bimal					
Chatur					
Total Score	15	15	16	16	18

It is known that Chatur always scores in multiples of 3. His score on Day 2 is the unique highest score in the competition. His minimum score is observed only on Day 1, and it matches Akhil's score on Day 4.

Hence, only Chatur scored 9 (one time) on Day 2, and no other person scored 9 on any of the given 5 days. Chatur scored 3 only one time, which was on Day 1. Therefore, the scores obtained by Chatur on Day 3, Day 4, and Day 5 are 6, 6, and 6, respectively. It is also known that Akhil's score on Day 4 is the same as the score obtained by Chatur on Day 1. Hence, Akhil's score on Day 4 is 3.

Hence, we get the following table:

	Day 1	Day 2	Day 3	Day 4	Day 5
Akhil				3	
Bimal				7	
Chatur	3	9	6	6	6
Total Score	15	15	16	16	18

From Table 2, we see that the rank of Bimal and Akhil is the same, which is 2. Hence, The score obtained by Akhil and Bimal is the same. Let the score be x . Therefore, $6 + 2x = 16 \Rightarrow x = 5$

The rank of Chatur on Day 5 is 2, and the rank of Bimal is 1, which implies the score obtained by Bimal will be more than Chatur. Hence, Bimal can score either 7 or 8 on Day 5. Therefore, the score obtained by Akhil on Day 5 is either 5 or 4.

	Day 1	Day 2	Day 3	Day 4	Day 5
Akhil			5	3	5/4
Bimal			5	7	7/8
Chatur	3	9	6	6	6
Total Score	15	15	16	16	18

It is given that Bimal's scores are the same on Day 1 and Day 3. Hence, the score obtained by Bimal on Day 1 is 5, which implies The score obtained by Akhil is 7 on Day 1.

From Table 2, we can see that the rank of Bimal is 3 on Day 2, and the rank of Akhil is 2 on Day 2. Hence, the score of Bimal will be lower than Akhil on Day 2.

Let the score of Akhil be a , and the score of Bimal be b . Then $9+a+b = 15$, and $a > b$

$\Rightarrow a+b = 6$, and $a > b$

Hence, the value of a can be $4/5$, and the value of b can be $2/1$

Therefore, the final table is given below:

	Day 1	Day 2	Day 3	Day 4	Day 5	Total score
Akhil	7	4/5	5	3	5/4	23/24/25
Bimal	5	2/1	5	7	7/8	27/26/25
Chatur	3	9	6	6	6	30
Total Score	15	15	16	16	18	80

In the question, it is given that the total score obtained by Bimal is a multiple of 3, which implies the total score obtained by Bimal is 27, which implies the total score obtained by Akhil is 23.

Akhil will score 23, when his scores on Days 1, 2, 3, 4, and 5 are 7, 4, 5, 3, 4, respectively.

Hence, the score obtained by him on Day 2 is 4.

The correct option is D

 VIDEO SOLUTION

100+ CAT VARC Questions



26. B

Table 1: 2-day averages for Days 2 through 5			
Day 2	Day 3	Day 4	Day 5
15	15.5	16	17

Let the total score of day 1, day 2, day 3, day 4, and day 5 are d_1 , d_2 , d_3 , d_4 , and d_5 , respectively.

The table shows that $d_1+d_2 = 30 \dots \text{eq (1)}$, $d_2+d_3 = 31 \dots \text{eq (2)}$, $d_3+d_4 = 32 \dots \text{eq(3)}$, $d_4+d_5 = 34 \dots \text{eq(4)}$

It is given that participants are ranked each day, with the person having the maximum score being awarded the minimum rank (1) on that day. All participants with a tied score are awarded the best available rank if there is a tie.

It is given that the total score on Day 3 is the same as the total score on Day 4.

Therefore, $d_3 = d_4 \Rightarrow d_3 = d_4 = 16$, which implies $d_2 = 15$, $d_5 = 18$, and $d_1 = 15$.

The day-wise score is given below:

	Day 1	Day 2	Day 3	Day 4	Day 5
Akhil					
Bimal					
Chatur					
Total Score	15	15	16	16	18

It is known that Chatur always scores in multiples of 3. His score on Day 2 is the unique highest score in the competition. His minimum score is observed only on Day 1, and it matches Akhil's score on Day 4.

Hence, only Chatur scored 9 (one time) on Day 2, and no other person scored 9 on any of the given 5 days. Chatur scored 3 only one time, which was on Day 1. Therefore, the scores obtained by Chatur on Day 3, Day 4, and Day 5 are 6, 6, and 6, respectively. It is also known that Akhil's score on Day 4 is the same as the score obtained by Chatur on Day 1. Hence, Akhil's score on Day 4 is 3.

Hence, we get the following table:

	Day 1	Day 2	Day 3	Day 4	Day 5
Akhil				3	
Bimal				7	
Chatur	3	9	6	6	6
Total Score	15	15	16	16	18

From Table 2, we see that the rank of Bimal and Akhil is the same, which is 2. Hence, The score obtained by Akhil and Bimal is the same. Let the score be x . Therefore, $6+2x = 16 \Rightarrow x = 5$

The rank of Chatur on Day 5 is 2, and the rank of Bimal is 1, which implies the score obtained by Bimal will be more than Chatur. Hence, Bimal can score either 7 or 8 on Day 5. Therefore, the score obtained by Akhil on Day 5 is either 5 or 4.

	Day 1	Day 2	Day 3	Day 4	Day 5
Akhil			5	3	5/4
Bimal			5	7	7/8
Chatur	3	9	6	6	6
Total Score	15	15	16	16	18

It is given that Bimal's scores are the same on Day 1 and Day 3. Hence, the score obtained by Bimal on Day 1 is 5, which implies The score obtained by Akhil is 7 on Day 1.

From Table 2, we can see that the rank of Bimal is 3 on Day 2, and the rank of Akhil is 2 on Day 2. Hence, the score of Bimal will be lower than Akhil on Day 2.

Let the score of Akhil be a , and the score of Bimal be b . Then $9+a+b = 15$, and $a > b$

$\Rightarrow a+b = 6$, and $a > b$

Hence, the value of a can be $4/5$, and the value of b can be $2/1$

Therefore, the final table is given below:

	Day 1	Day 2	Day 3	Day 4	Day 5
Akhil	7	4/5	5	3	5/4
Bimal	5	2/1	5	7	7/8
Chatur	3	9	6	6	6
Total Score	15	15	16	16	18

From the table, we can see that the score of Akhil is 7 on day 1.

The correct option is B

VIDEO SOLUTION

27.D

Table 1: 2-day averages for Days 2 through 5			
Day 2	Day 3	Day 4	Day 5
15	15.5	16	17

Let the total score of day 1, day 2, day 3, day 4, and day 5 are d_1 , d_2 , d_3 , d_4 , and d_5 , respectively.

The table shows that $d_1 + d_2 = 30 \dots \text{eq (1)}$, $d_2 + d_3 = 31 \dots \text{eq (2)}$, $d_3 + d_4 = 32 \dots \text{eq (3)}$, $d_4 + d_5 = 34 \dots \text{eq (4)}$

It is given that participants are ranked each day, with the person having the maximum score being awarded the minimum rank (1) on that day. All participants with a tied score are awarded the best available rank if there is a tie.

It is given that the total score on Day 3 is the same as the total score on Day 4.

Therefore, $d_3 = d_4 \Rightarrow d_3 = d_4 = 16$, which implies $d_2 = 15$, $d_5 = 18$, and $d_1 = 15$.

The day-wise score is given below:

	Day 1	Day 2	Day 3	Day 4	Day 5
Akhil					
Bimal					
Chatur					
Total Score	15	15	16	16	18

It is known that Chatur always scores in multiples of 3. His score on Day 2 is the unique highest score in the competition. His minimum score is observed only on Day 1, and it matches Akhil's score on Day 4.

Hence, only Chatur scored 9 (one time) on Day 2, and no other person scored 9 on any of the given 5 days. Chatur scored 3 only one time, which was on Day 1. Therefore, the scores obtained by Chatur on Day 3, Day 4, and Day 5 are 6, 6, and 6, respectively. It is also known that Akhil's score on Day 4 is the same as the score obtained by Chatur on Day 1. Hence, Akhil's score on Day 4 is 3.

Hence, we get the following table:

	Day 1	Day 2	Day 3	Day 4	Day 5
Akhil				3	
Bimal				7	
Chatur	3	9	6	6	6
Total Score	15	15	16	16	18

From Table 2, we see that the rank of Bimal and Akhil is the same, which is 2. Hence, The score obtained by Akhil and Bimal is the same. Let the score be x . Therefore, $6 + 2x = 16 \Rightarrow x = 5$

The rank of Chatur on Day 5 is 2, and the rank of Bimal is 1, which implies the score obtained by Bimal will be more than Chatur. Hence, Bimal can score either 7 or 8 on Day 5. Therefore, the score obtained by Akhil on Day 5 is either 5 or 4.

	Day 1	Day 2	Day 3	Day 4	Day 5
Akhil			5	3	5/4
Bimal			5	7	7/8
Chatur	3	9	6	6	6
Total Score	15	15	16	16	18

It is given that Bimal's scores are the same on Day 1 and Day 3. Hence, the score obtained by Bimal on Day 1 is 5, which implies The score obtained by Akhil is 7 on Day 1.

From Table 2, we can see that the rank of Bimal is 3 on Day 2, and the rank of Akhil is 2 on Day 2. Hence, the score of Bimal will be lower than Akhil's on Day 2.

Let the score of Akhil be a , and the score of Bimal be b . Then $9+a+b = 15$, and $a > b$

$\Rightarrow a+b=6$, and $a > b$

Hence, the value of a can be $4/5$, and the value of b can be $2/1$

Therefore, the final table is given below:

	Day 1	Day 2	Day 3	Day 4	Day 5	Total score
Akhil	7	4/5	5	3	5/4	23/24/25
Bimal	5	2/1	5	7	7/8	27/26/25
Chatur	3	9	6	6	6	30
Total Score	15	15	16	16	18	80

From the table, we can see that the maximum score is obtained by Chatur.

The correct option is D

 VIDEO SOLUTION

28. 1500

Let us assume units of Aurels, Brins, Crowns, and Zentars to be A, B, C and D, respectively.

We are given the travel cost of Jano to be $3500A$ in Statement 1.

We are also given the travel cost of Kira to be $8000A$ in Statement 1 and $4000B$ in Statement 2.

We are also given the travel cost of Lian to be $36000C$ in statement 3.

We know that the travel cost has to be constant throughout, so by equating the travel costs of Kira, we get,

$$8000A = 4000B$$

$$B = 2A \text{ ---(1)}$$

We are also given the value of C to be $0.5Z$.

$$C = 0.5Z \text{ ---(2)}$$

$$\text{So, the travel cost of Lian} = 36000C = 36000 \times 0.5Z = 18000Z$$

Let us put the known information in the table, and we get,

TRAVELLER	TRAVEL COST	SPENDING COST		FLIGHT COST
		COUNTRY	MONEY SPENT	
JANO	3500A	Aurevia		4000Z
		Cyrenia		
KIRA	8000A	Aurevia		
		Brelosia	2000B	
LIAN	18000Z	Brelosia	3000B	
		Cyrenia		

We are given that the spending cost of any individual is different in different cities.

We are given that the spending amounts are amongst $\{1000, 2000, 3000\}$ in their local currencies and also told that there were two spends of 1000 and one spend of 3000, which makes the total spending amount 2000, to be 3 because there are 6 spending amounts in total.

One 2000 and one 3000 are already assigned in the above table, so we will be left with two spends of 1000 and two spends of 2000 to be assigned.

Kira has already spent 2000, so the only amount that she must have spent is 1000, as she cannot spend 2000 twice in different countries.

Lian has already spent 3000, so the only amount that she must have spent is 2000. If she spent 1000, then Jano will be left with two 2000s, which is not possible.

We can also conclude that Jano spend 1000 and 2000 in some order in the two countries.

The flight costs are given as 4000Z, 5000Z, and 6000Z, out of which 4000Z is for Jano and 5000Z and 6000Z are for Kira and Lian, in some order.

Filling up the table with these values, we get,

TRAVELLER	TRAVEL COST	SPENDING COST		FLIGHT COST
		COUNTRY	MONEY SPENT	
JANO	3500A	Aurevia	1000A/2000A	4000Z
		Cyrenia	2000C/1000C	
KIRA	8000A	Aurevia	1000A	5000Z/6000Z
		Brelosia	2000B = 4000A	
LIAN	18000Z	Brelosia	3000B = 6000A	6000Z/5000Z
		Cyrenia	2000C = 1000Z	

Travel Cost = Spending Cost + Flight Cost

For Kira,

Travel Cost = 8000A

Spending Cost = 1000A + 4000A = 5000A

Flight Cost = 5000Z/6000Z

CASE 1: Flight cost of Kira = 5000Z and Flight cost of Lian = 6000Z

If we assume the Flight cost to be 5000Z, then we get,

$8000A = 5000A + 5000Z$

$3000A = 5000Z$

$A = \frac{5}{3}Z$

For Lian,

Travel Cost = 18000Z

Spending Cost = 6000A + 1000Z = $6000 \times \frac{5}{3}Z + 1000Z = 11000Z$

Flight Cost = Travel Cost - Spending Cost = 18000Z - 11000Z = 7000Z

But in our assumption, the Flight Cost of Lian is 6000Z, which does not match the above answer.

So, we can eliminate this case.

CASE 2: Flight cost of Kira = 6000Z and Flight cost of Lian = 5000Z

If we assume the Flight cost to be 6000Z, then we get,

$8000A = 5000A + 6000Z$

$3000A = 6000Z$

$A = 2Z$

For Lian,

Travel Cost = 18000Z

Spending Cost = 6000A + 1000Z = $6000 \times 2Z + 1000Z = 13000Z$

Flight Cost = Travel Cost - Spending Cost = 18000Z - 13000Z = 5000Z

In our assumption, the Flight Cost of Lian is 5000Z, which matches the above answer.

So, Flight cost of Kira = 6000Z, Flight cost of Lian = 5000Z and A = 2Z.

Filling the table with calculated values all in the currency of Z, we get,

TRAVELLER	TRAVEL COST	SPENDING COST		FLIGHT COST
		COUNTRY	MONEY SPENT	
JANO	7000Z	Aurevia	2000Z/4000Z	4000Z
		Cyrenia	1000Z/500Z	
KIRA	16000Z	Aurevia	2000Z	6000Z
		Brelosia	8000Z	
LIAN	18000Z	Brelosia	12000Z	5000Z
		Cyrenia	1000Z	

In the case of Zano,

Spending Cost = Travel Cost - Flight Cost = 7000Z - 4000Z = 3000Z

In the first case, the Spending Cost = 2000Z + 1000Z = 3000Z

In the second case, the Spending Cost = 4000Z + 500Z = 4500Z

We obtained a spending cost of 3000Z only in the first case, so we can eliminate the second case.

The final table looks like,

TRAVELLER	TRAVEL COST	SPENDING COST		FLIGHT COST
		COUNTRY	MONEY SPENT	
JANO	7000Z	Aurevia	2000Z	4000Z
		Cyrenia	1000Z	
KIRA	16000Z	Aurevia	2000Z	6000Z
		Brelosia	8000Z	
LIAN	18000Z	Brelosia	12000Z	5000Z
		Cyrenia	1000Z	

Total Spending Cost of Jano = 2000Z + 1000Z = 3000Z

We know that A = 2Z, so the above value in A can be calculated as,

$$3000Z = 3000 \cdot A/2 = 1500A$$

So, the spending of Jano in Aurels is 1500A.

Hence, the correct answer is 1500.

[VIEW SOLUTION](#)

29. A

Let us assume units of Aurels, Brins, Crowns, and Zentars to be A, B, C and D, respectively.

We are given the travel cost of Jano to be 3500A in Statement 1.

We are also given the travel cost of Kira to be 8000A in Statement 1 and 4000B in Statement 2.

We are also given the travel cost of Lian to be 36000C in statement 3.

We know that the travel cost has to be constant throughout, so by equating the travel costs of Kira, we get,

$$8000A = 4000B$$

$$B = 2A \text{ ---(1)}$$

We are also given the value of C to be 0.5 Z.

$$C = 0.5Z \text{ ---(2)}$$

$$\text{So, the travel cost of Lian} = 36000C = 36000 \cdot 0.5Z = 18000Z$$

Let us put the known information in the table, and we get,

TRAVELLER	TRAVEL COST	SPENDING COST		FLIGHT COST
		COUNTRY	MONEY SPENT	
JANO	3500A	Aurevia		4000Z
		Cyrenia		
KIRA	8000A	Aurevia		
		Brelosia	2000B	
LIAN	18000Z	Brelosia	3000B	
		Cyrenia		

We are given that the spending cost of any individual is different in different cities.

We are given that the spending amounts are amongst {1000, 2000, 3000} in their local currencies and also told that there were two spends of 1000 and one spend of 3000, which makes the total spending amount 2000, to be 3 because there are 6 spending amounts in total.

One 2000 and one 3000 are already assigned in the above table, so we will be left with two spends of 1000 and two spends of 2000 to be assigned.

Kira has already spent 2000, so the only amount that she must have spent is 1000, as she cannot spend 2000 twice in different countries.

Lian has already spent 3000, so the only amount that she must have spent is 2000. If she spent 1000, then Jano will be left with two 2000s, which is not possible.

We can also conclude that Jano spent 1000 and 2000 in some order in the two countries.

The flight costs are given as 4000Z, 5000Z, and 6000Z, out of which 4000Z is for Jano and 5000Z and 6000Z are for Kira and Lian, in some order.

Filling up the table with these values, we get,

TRAVELLER	TRAVEL COST	SPENDING COST		FLIGHT COST
		COUNTRY	MONEY SPENT	
JANO	3500A	Aurevia	1000A/2000A	4000Z
		Cyrenia	2000C/1000C	
KIRA	8000A	Aurevia	1000A	5000Z/6000Z
		Brelosia	2000B = 4000A	
LIAN	18000Z	Brelosia	3000B = 6000A	6000Z/5000Z
		Cyrenia	2000C = 1000Z	

Travel Cost = Spending Cost + Flight Cost

For Kira,

Travel Cost = 8000A

Spending Cost = 1000A + 4000A = 5000A

Flight Cost = 5000Z/6000Z

CASE 1: Flight cost of Kira = 5000Z and Flight cost of Lian = 6000Z

If we assume the Flight cost to be 5000Z, then we get,

$8000A = 5000A + 5000Z$

$3000A = 5000Z$

$A = \frac{5}{3}Z$

For Lian,

Travel Cost = 18000Z

Spending Cost = 6000A + 1000Z = $6000 \times \frac{5}{3}Z + 1000Z = 11000Z$

Flight Cost = Travel Cost - Spending Cost = $18000Z - 11000Z = 7000Z$

But in our assumption, the Flight Cost of Lian is 6000Z, which does not match the above answer.

So, we can eliminate this case.

CASE 2: Flight cost of Kira = 6000Z and Flight cost of Lian = 5000Z

If we assume the Flight cost to be 6000Z, then we get,

$$8000A = 5000A + 6000Z$$

$$3000A = 6000Z$$

$$A = 2Z$$

For Lian,

$$\text{Travel Cost} = 18000Z$$

$$\text{Spending Cost} = 6000A + 1000Z = 6000 \cdot 2Z + 1000Z = 13000Z$$

$$\text{Flight Cost} = \text{Travel Cost} - \text{Spending Cost} = 18000Z - 13000Z = 5000Z$$

In our assumption, the Flight Cost of Lian is 5000Z, which matches the above answer.

So, Flight cost of Kira = 6000Z, Flight cost of Lian = 5000Z and $A = 2Z$.

Filling the table with calculated values all in the currency of Z, we get,

TRAVELLER	TRAVEL COST	SPENDING COST		FLIGHT COST
		COUNTRY	MONEY SPENT	
JANO	7000Z	Aurevia	2000Z/4000Z	4000Z
		Cyrenia	1000Z/500Z	
KIRA	16000Z	Aurevia	2000Z	6000Z
		Brelosia	8000Z	
LIAN	18000Z	Brelosia	12000Z	5000Z
		Cyrenia	1000Z	

In the case of Zano,

$$\text{Spending Cost} = \text{Travel Cost} - \text{Flight Cost} = 7000Z - 4000Z = 3000Z$$

$$\text{In the first case, the Spending Cost} = 2000Z + 1000Z = 3000Z$$

$$\text{In the second case, the Spending Cost} = 4000Z + 500Z = 4500Z$$

We obtained a spending cost of 3000Z only in the first case, so we can eliminate the second case.

The final table looks like,

TRAVELLER	TRAVEL COST	SPENDING COST		FLIGHT COST
		COUNTRY	MONEY SPENT	
JANO	7000Z	Aurevia	2000Z	4000Z
		Cyrenia	1000Z	
KIRA	16000Z	Aurevia	2000Z	6000Z
		Brelosia	8000Z	
LIAN	18000Z	Brelosia	12000Z	5000Z
		Cyrenia	1000Z	

We know that $B = 2A$ and $A = 2Z$, so the value of B in terms of Z is

$$B = 2 \cdot 2Z = 4Z \quad \text{--(3)}$$

We also know that

$$C = 0.5Z \quad \text{--(4)}$$

Dividing (3) by (4), we get,

$$\frac{B}{C} = \frac{4Z}{0.5Z} = 8$$

$$B = 8C$$

So, one Brin is equivalent to 8 crowns.

Hence, the correct answer is option A.

[VIEW SOLUTION](#)

30. 13000

Let us assume units of Aurels, Brins, Crowns, and Zentars to be A, B, C and D, respectively.

We are given the travel cost of Jano to be 3500A in Statement 1.

We are also given the travel cost of Kira to be 8000A in Statement 1 and 4000B in Statement 2.

We are also given the travel cost of Lian to be 36000C in statement 3.

We know that the travel cost has to be constant throughout, so by equating the travel costs of Kira, we get,
 $8000A = 4000B$

$$B = 2A \text{ ---(1)}$$

We are also given the value of C to be 0.5 Z.

$$C = 0.5Z \text{ ---(2)}$$

So, the travel cost of Lian = $36000C = 36000 \times 0.5Z = 18000Z$

Let us put the known information in the table, and we get,

TRAVELLER	TRAVEL COST	SPENDING COST		FLIGHT COST
		COUNTRY	MONEY SPENT	
JANO	3500A	Aurevia		4000Z
		Cyrenia		
KIRA	8000A	Aurevia		
		Brelosia	2000B	
LIAN	18000Z	Brelosia	3000B	
		Cyrenia		

We are given that the spending cost of any individual is different in different cities.

We are given that the spending amounts are amongst {1000, 2000, 3000} in their local currencies and also told that there were two spends of 1000 and one spend of 3000, which makes the total spending amount 2000, to be 3 because there are 6 spending amounts in total.

One 2000 and one 3000 are already assigned in the above table, so we will be left with two spends of 1000 and two spends of 2000 to be assigned.

Kira has already spent 2000, so the only amount that she must have spent is 1000, as she cannot spend 2000 twice in different countries.

Lian has already spent 3000, so the only amount that she must have spent is 2000. If she spent 1000, then Jano will be left with two 2000s, which is not possible.

We can also conclude that Jano spent 1000 and 2000 in some order in the two countries.

The flight costs are given as 4000Z, 5000Z, and 6000Z, out of which 4000Z is for Jano and 5000Z and 6000Z are for Kira and Lian, in some order.

Filling up the table with these values, we get,

TRAVELLER	TRAVEL COST	SPENDING COST		FLIGHT COST
		COUNTRY	MONEY SPENT	
JANO	3500A	Aurevia	1000A/2000A	4000Z
		Cyrenia	2000C/1000C	
KIRA	8000A	Aurevia	1000A	5000Z/6000Z
		Brelosia	2000B = 4000A	
LIAN	18000Z	Brelosia	3000B = 6000A	6000Z/5000Z
		Cyrenia	2000C = 1000Z	

Travel Cost = Spending Cost + Flight Cost

For Kira,

Travel Cost = 8000A

Spending Cost = 1000A + 4000A = 5000A

Flight Cost = 5000Z/6000Z

CASE 1: Flight cost of Kira = 5000Z and Flight cost of Lian = 6000Z

If we assume the Flight cost to be 5000Z, then we get,

$$8000A = 5000A + 5000Z$$

$$3000A = 5000Z$$

$$A = \frac{5}{3}Z$$

For Lian,

Travel Cost = 18000Z

$$\text{Spending Cost} = 6000A + 1000Z = 6000 \times \frac{5}{3}Z + 1000Z = 11000Z$$

$$\text{Flight Cost} = \text{Travel Cost} - \text{Spending Cost} = 18000Z - 11000Z = 7000Z$$

But in our assumption, the Flight Cost of Lian is 6000Z, which does not match the above answer.

So, we can eliminate this case.

CASE 2: Flight cost of Kira = 6000Z and Flight cost of Lian = 5000Z

If we assume the Flight cost to be 6000Z, then we get,

$$8000A = 5000A + 6000Z$$

$$3000A = 6000Z$$

$$A = 2Z$$

For Lian,

Travel Cost = 18000Z

$$\text{Spending Cost} = 6000A + 1000Z = 6000 \times 2Z + 1000Z = 13000Z$$

$$\text{Flight Cost} = \text{Travel Cost} - \text{Spending Cost} = 18000Z - 13000Z = 5000Z$$

In our assumption, the Flight Cost of Lian is 5000Z, which matches the above answer.

So, Flight cost of Kira = 6000Z, Flight cost of Lian = 5000Z and A = 2Z.

Filling the table with calculated values all in the currency of Z, we get,

TRAVELLER	TRAVEL COST	SPENDING COST		FLIGHT COST
		COUNTRY	MONEY SPENT	
JANO	7000Z	Aurevia	2000Z/4000Z	4000Z
		Cyrenia	1000Z/500Z	
KIRA	16000Z	Aurevia	2000Z	6000Z
		Brelosia	8000Z	
LIAN	18000Z	Brelosia	12000Z	5000Z
		Cyrenia	1000Z	

In the case of Zano,

$$\text{Spending Cost} = \text{Travel Cost} - \text{Flight Cost} = 7000Z - 4000Z = 3000Z$$

$$\text{In the first case, the Spending Cost} = 2000Z + 1000Z = 3000Z$$

$$\text{In the second case, the Spending Cost} = 4000Z + 500Z = 4500Z$$

We obtained a spending cost of 3000Z only in the first case, so we can eliminate the second case.

The final table looks like,

TRAVELLER	TRAVEL COST	SPENDING COST		FLIGHT COST
		COUNTRY	MONEY SPENT	
JANO	7000Z	Aurevia	2000Z	4000Z
		Cyrenia	1000Z	
KIRA	16000Z	Aurevia	2000Z	6000Z
		Brelosia	8000Z	
LIAN	18000Z	Brelosia	12000Z	5000Z
		Cyrenia	1000Z	

Spending Cost of Lian = 12000Z + 1000Z = 13000Z

Hence, the correct answer is 13000.

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31. A

Let us assume units of Aurels, Brins, Crowns, and Zentars to be A, B, C and D, respectively.

We are given the travel cost of Jano to be 3500A in Statement 1.

We are also given the travel cost of Kira to be 8000A in Statement 1 and 4000B in Statement 2.

We are also given the travel cost of Lian to be 36000C in statement 3.

We know that the travel cost has to be constant throughout, so by equating the travel costs of Kira, we get,
 $8000A = 4000B$

$$B = 2A \text{ ---(1)}$$

We are also given the value of C to be 0.5 Z.

$$C = 0.5Z \text{ ---(2)}$$

So, the travel cost of Lian = 36000C = 36000*0.5Z = 18000Z

Let us put the known information in the table, and we get,

TRAVELLER	TRAVEL COST	SPENDING COST		FLIGHT COST
		COUNTRY	MONEY SPENT	
JANO	3500A	Aurevia		4000Z
		Cyrenia		
KIRA	8000A	Aurevia		
		Brelosia	2000B	
LIAN	18000Z	Brelosia	3000B	
		Cyrenia		

We are given that the spending cost of any individual is different in different cities.

We are given that the spending amounts are amongst {1000, 2000, 3000} in their local currencies and also told that there were two spends of 1000 and one spend of 3000, which makes the total spending amount 2000, to be 3 because there are 6 spending amounts in total.

One 2000 and one 3000 are already assigned in the above table, so we will be left with two spends of 1000 and two spends of 2000 to be assigned.

Kira has already spent 2000, so the only amount that she must have spent is 1000, as she cannot spend 2000 twice in different countries.

Lian has already spent 3000, so the only amount that she must have spent is 2000. If she spent 1000, then Jano will be left with two 2000s, which is not possible.

We can also conclude that Jano spend 1000 and 2000 in some order in the two countries.

The flight costs are given as 4000Z, 5000Z, and 6000Z, out of which 4000Z is for Jano and 5000Z and 6000Z are for Kira and Lian, in some order.

Filling up the table with these values, we get,

TRAVELLER	TRAVEL COST	SPENDING COST		FLIGHT COST
		COUNTRY	MONEY SPENT	
JANO	3500A	Aurevia	1000A/2000A	4000Z
		Cyrenia	2000C/1000C	
KIRA	8000A	Aurevia	1000A	5000Z/6000Z
		Brelosia	2000B = 4000A	
LIAN	18000Z	Brelosia	3000B = 6000A	6000Z/5000Z
		Cyrenia	2000C = 1000Z	

Travel Cost = Spending Cost + Flight Cost

For Kira,

Travel Cost = 8000A

Spending Cost = 1000A + 4000A = 5000A

Flight Cost = 5000Z/6000Z

CASE 1: Flight cost of Kira = 5000Z and Flight cost of Lian = 6000Z

If we assume the Flight cost to be 5000Z, then we get,

$8000A = 5000A + 5000Z$

$3000A = 5000Z$

$A = \frac{5}{3}Z$

For Lian,

Travel Cost = 18000Z

Spending Cost = 6000A + 1000Z = $6000 \times \frac{5}{3}Z + 1000Z = 11000Z$

Flight Cost = Travel Cost - Spending Cost = $18000Z - 11000Z = 7000Z$

But in our assumption, the Flight Cost of Lian is 6000Z, which does not match the above answer.

So, we can eliminate this case.

CASE 2: Flight cost of Kira = 6000Z and Flight cost of Lian = 5000Z

If we assume the Flight cost to be 6000Z, then we get,

$8000A = 5000A + 6000Z$

$3000A = 6000Z$

$A = 2Z$

For Lian,

Travel Cost = 18000Z

Spending Cost = 6000A + 1000Z = $6000 \times 2Z + 1000Z = 13000Z$

Flight Cost = Travel Cost - Spending Cost = $18000Z - 13000Z = 5000Z$

In our assumption, the Flight Cost of Lian is 5000Z, which matches the above answer.

So, Flight cost of Kira = 6000Z, Flight cost of Lian = 5000Z and $A = 2Z$.

Filling the table with calculated values all in the currency of Z, we get,

TRAVELLER	TRAVEL COST	SPENDING COST		FLIGHT COST
		COUNTRY	MONEY SPENT	
JANO	7000Z	Aurevia	2000Z/4000Z	4000Z
		Cyrenia	1000Z/500Z	
KIRA	16000Z	Aurevia	2000Z	6000Z
		Brelosia	8000Z	
LIAN	18000Z	Brelosia	12000Z	5000Z
		Cyrenia	1000Z	

In the case of Zano,

Spending Cost = Travel Cost - Flight Cost = 7000Z - 4000Z = 3000Z

In the first case, the Spending Cost = 2000Z + 1000Z = 3000Z

In the second case, the Spending Cost = 4000Z + 500Z = 4500Z

We obtained a spending cost of 3000Z only in the first case, so we can eliminate the second case.

The final table looks like,

TRAVELLER	TRAVEL COST	SPENDING COST		FLIGHT COST
		COUNTRY	MONEY SPENT	
JANO	7000Z	Aurevia	2000Z	4000Z
		Cyrenia	1000Z	
KIRA	16000Z	Aurevia	2000Z	6000Z
		Brelosia	8000Z	
LIAN	18000Z	Brelosia	12000Z	5000Z
		Cyrenia	1000Z	

Statement 1) Jano spent 2000 in Aurevia

Jano spent 2000Z in Aurevia = 1000A in Aurevia.

So, this option is incorrect.

Statement 2) Lian spent 2000 in Cyrenia

Lian spent 1000Z in Cyrenia = 2000C in Cyrenia.

So, this option is correct.

Statement 3) Jano spent 2000 in Cyrenia

Jano spent 1000Z in Cyrenia = 2000C in Cyrenia.

So, this option is correct.

Statement 4) Kira spent 1000 in Aurevia

Kira spent 2000Z in Aurevia = 1000A in Aurevia.

So, this option is correct.

Hence, the correct answer is option A.

[VIEW SOLUTION](#)

32. 41000

Let us assume units of Aurels, Brins, Crowns, and Zentars to be A, B, C and D, respectively.

We are given the travel cost of Jano to be 3500A in Statement 1.

We are also given the travel cost of Kira to be 8000A in Statement 1 and 4000B in Statement 2.

We are also given the travel cost of Lian to be 36000C in statement 3.

We know that the travel cost has to be constant throughout, so by equating the travel costs of Kira, we get,

$8000A = 4000B$

$$B = 2A \text{ ---(1)}$$

We are also given the value of C to be 0.5 Z.

$$C = 0.5Z \text{ ---(2)}$$

So, the travel cost of Lian = $36000C = 36000 \times 0.5Z = 18000Z$

Let us put the known information in the table, and we get,

TRAVELLER	TRAVEL COST	SPENDING COST		FLIGHT COST
		COUNTRY	MONEY SPENT	
JANO	3500A	Aurevia		4000Z
		Cyrenia		
KIRA	8000A	Aurevia		
		Brelosia	2000B	
LIAN	18000Z	Brelosia	3000B	
		Cyrenia		

We are given that the spending cost of any individual is different in different cities.

We are given that the spending amounts are amongst {1000, 2000, 3000} in their local currencies and also told that there were two spends of 1000 and one spend of 3000, which makes the total spending amount 2000, to be 3 because there are 6 spending amounts in total.

One 2000 and one 3000 are already assigned in the above table, so we will be left with two spends of 1000 and two spends of 2000 to be assigned.

Kira has already spent 2000, so the only amount that she must have spent is 1000, as she cannot spend 2000 twice in different countries.

Lian has already spent 3000, so the only amount that she must have spent is 2000. If she spent 1000, then Jano will be left with two 2000s, which is not possible.

We can also conclude that Jano spend 1000 and 2000 in some order in the two countries.

The flight costs are given as 4000Z, 5000Z, and 6000Z, out of which 4000Z is for Jano and 5000Z and 6000Z are for Kira and Lian, in some order.

Filling up the table with these values, we get,

TRAVELLER	TRAVEL COST	SPENDING COST		FLIGHT COST
		COUNTRY	MONEY SPENT	
JANO	3500A	Aurevia	1000A/2000A	4000Z
		Cyrenia	2000C/1000C	
KIRA	8000A	Aurevia	1000A	5000Z/6000Z
		Brelosia	2000B = 4000A	
LIAN	18000Z	Brelosia	3000B = 6000A	6000Z/5000Z
		Cyrenia	2000C = 1000Z	

Travel Cost = Spending Cost + Flight Cost

For Kira,

Travel Cost = 8000A

Spending Cost = 1000A + 4000A = 5000A

Flight Cost = 5000Z/6000Z

CASE 1: Flight cost of Kira = 5000Z and Flight cost of Lian = 6000Z

If we assume the Flight cost to be 5000Z, then we get,

$$8000A = 5000A + 5000Z$$

$$3000A = 5000Z$$

$$A = \frac{5}{3}Z$$

33.14

Let us assume, A is the total number of AC's sold

=> From the information that the total number of ACs sold in the city, 25% were of Window variant => Window AC's = $A/4$ and Split AC's = $3A/4$

Now, let us assume B is the total number of inverter ACs

=> From the information that among the Inverter ACs sold, 20% were of Window variant. => Window Inverter AC's = $B/5$ and Window Non-Inverter AC's = $4B/5$

Total (A)			
Split ($3A/4$)		Window ($A/4$)	
Inv ($4B/5$)	Non-Inv	Inv ($B/5$)	Non-Inv

From - Condition-3

=> $A/4 - B/5 = 6$ and $4B/5 = 36$ => $B = 46$ and $A = 60$.

Total = 60			
Split = 45		Window = 15	
Inv = 36	Non-Inv = 9	Inv = 9	Non-Inv = 6

Now, from condition-6

a) D1 & D4 sold "0" window Non-inverter ACs => D2 & D3 sold 6 window non-inverter ACs, it is given that D2 sold twice as many as D3 => D2 sold 4 and D3 sold 2 ACs of this type.

From condition-2

b) Let us assume, D1 sold "x" window inverter ACs => Number of split inverter ACs sold is $13-x$

From condition-4

c) Number of split ACs sold by D1 will be "2x"

From condition-5

d) Let us assume 'y' is the number of window ACs sold by D3 & D4 => D2 sold $3y$ ACs of this type.

From condition-7

e) Let us assume 'z' is the number of split inverter ACs sold by D3 and D4 => D2 sold $2z$ ACs of this type.

Let us use a, b, c, d, and e make a table:

D1 Total =			
Split =		Window = x	
Inv = $13-x$	Non-Inv =	Inv = x	Non-Inv = 0

D2 Total =			
Split =		Window = $3y$	
Inv = $2z$	Non-Inv =	Inv =	Non-Inv = 4

D3 Total =			
Split =		Window = y	
Inv = z	Non-Inv = 3	Inv =	Non-Inv = 2

D4 Total =			
Split =		Window = y	
Inv = z	Non-Inv =	Inv =	Non-Inv = 0

We know that the total number of window ACs is 15

=> $x + 3y + y + y = 15$ => $x + 5y = 15$, also x and y should be greater than or equal to 2 from condition-1

=> $x = 5$ and $y = 2$ is the only solution.

Filling this in the table:

D1 Total =			
Split =		Window = 5	
Inv = 8	Non-Inv =	Inv = 5	Non-Inv = 0

D2 Total =			
Split =		Window = 6	
Inv = 2z	Non-Inv =	Inv = 2	Non-Inv = 4

D3 Total =			
Split =		Window = 2	
Inv = z	Non-Inv = 3	Inv = 0	Non-Inv = 2

D4 Total =			
Split =		Window = 2	
Inv = z	Non-Inv =	Inv = 2	Non-Inv = 0

Now, Number of split inverter ACs is 36

$$\Rightarrow 8 + 2z + z + z = 36 \Rightarrow 4z = 28 \Rightarrow z = 7.$$

Filling this and using (5), the number of split AC's sold by D1 is $2 \times 5 = 10$.

D1 Total = 15			
Split = 10		Window = 5	
Inv = 8	Non-Inv = 2	Inv = 5	Non-Inv = 0

D2 Total =			
Split =		Window = 6	
Inv = 14	Non-Inv =	Inv = 2	Non-Inv = 4

D3 Total = 12			
Split = 10		Window = 2	
Inv = 7	Non-Inv = 3	Inv = 0	Non-Inv = 2

D4 Total =			
Split =		Window = 2	
Inv = 7	Non-Inv =	Inv = 2	Non-Inv = 0

From the table, we see that 14 split inverter ACs are sold.

 VIDEO SOLUTION

34. D

Let us assume, A is the total number of AC's sold

$\Rightarrow$ From the information that the total number of ACs sold in the city, 25% were of Window variant $\Rightarrow$ Window AC's = $A/4$ and Split AC's = $3A/4$

Now, let us assume B is the total number of inverter ACs

$\Rightarrow$ From the information that among the Inverter ACs sold, 20% were of Window variant. $\Rightarrow$ Window Inverter AC's = $B/5$ and Window Non-Inverter AC's = $4B/5$

Total (A)			
Split ($3A/4$)		Window ($A/4$)	
Inv ($4B/5$)	Non-Inv	Inv ($B/5$)	Non-Inv

From - Condition-3

$$\Rightarrow A/4 - B/5 = 6 \text{ and } 4B/5 = 36 \Rightarrow B = 46 \text{ and } A = 60.$$

Total = 60			
Split = 45		Window = 15	
Inv = 36	Non-Inv = 9	Inv = 9	Non-Inv = 6

Now, from condition-6

a) D1 & D4 sold "0" window Non-inverter ACs => D2 & D3 sold 6 window non-inverter ACs, it is given that D2 sold twice as many as D3 => D2 sold 4 and D3 sold 2 ACs of this type.

From condition-2

b) Let us assume, D1 sold "x" window inverter ACs => Number of split inverter ACs sold is $13-x$

From condition-4

c) Number of split ACs sold by D1 will be "2x"

From condition-5

d) Let us assume 'y' is the number of window ACs sold by D3 & D4 => D2 sold $3y$ ACs of this type.

From condition-7

e) Let us assume 'z' is the number of split inverter ACs sold by D3 and D4 => D2 sold $2z$ ACs of this type.

Let us use a, b, c, d, and e make a table:

D1 Total =			
Split =		Window = x	
Inv = $13-x$	Non-Inv =	Inv = x	Non-Inv = 0

D2 Total =			
Split =		Window = $3y$	
Inv = $2z$	Non-Inv =	Inv =	Non-Inv = 4

D3 Total =			
Split =		Window = y	
Inv = z	Non-Inv = 3	Inv =	Non-Inv = 2

D4 Total =			
Split =		Window = y	
Inv = z	Non-Inv =	Inv =	Non-Inv = 0

We know that the total number of window ACs is 15

=> $x + 3y + y + y = 15$ => $x + 5y = 15$, also x and y should be greater than or equal to 2 from condition-1

=> $x = 5$ and $y = 2$ is the only solution.

Filling this in the table:

D1 Total =			
Split =		Window = 5	
Inv = 8	Non-Inv =	Inv = 5	Non-Inv = 0

D2 Total =			
Split =		Window = 6	
Inv = $2z$	Non-Inv =	Inv = 2	Non-Inv = 4

D3 Total =			
Split =		Window = 2	
Inv = z	Non-Inv = 3	Inv = 0	Non-Inv = 2

D4 Total =			
Split =		Window = 2	
Inv = z	Non-Inv =	Inv = 2	Non-Inv = 0

Now, Number of split inverter ACs is 36

=> $8 + 2z + z + z = 36$ => $4z = 28$ => $z = 7$.

Filling this and using (5), the number of split AC's sold by D1 is $2*5 = 10$.

D1 Total = 15			
Split = 10		Window = 5	
Inv = 8	Non-Inv = 2	Inv = 5	Non-Inv = 0

D2 Total =			
Split =		Window = 6	
Inv = 14	Non-Inv =	Inv = 2	Non-Inv = 4

D3 Total = 12			
Split = 10		Window = 2	
Inv = 7	Non-Inv = 3	Inv = 0	Non-Inv = 2

D4 Total =			
Split =		Window = 2	
Inv = 7	Non-Inv =	Inv = 2	Non-Inv = 0

We see that D1 & D3 sold 27 ACs together which is less than $60 - 27 = 33$ sold by D2 & D4.

=> Option-D is definitely false.

 VIDEO SOLUTION

35. B

Let us assume, A is the total number of AC's sold

=> From the information that the total number of ACs sold in the city, 25% were of Window variant => Window AC's = $A/4$ and Split AC's = $3A/4$

Now, let us assume B is the total number of inverter ACs

=> From the information that among the Inverter ACs sold, 20% were of Window variant. => Window Inverter AC's = $B/5$ and Window Non-Inverter AC's = $4B/5$

Total (A)			
Split ($3A/4$)		Window ($A/4$)	
Inv ($4B/5$)	Non-Inv	Inv ($B/5$)	Non-Inv

From - Condition-3

=> $A/4 - B/5 = 6$ and $4B/5 = 36$ => $B = 46$ and $A = 60$.

Total = 60			
Split = 45		Window = 15	
Inv = 36	Non-Inv = 9	Inv = 9	Non-Inv = 6

Now, from condition-6

a) D1 & D4 sold "0" window Non-inverter ACs => D2 & D3 sold 6 window non-inverter ACs, it is given that D2 sold twice as many as D3 => D2 sold 4 and D3 sold 2 ACs of this type.

From condition-2

b) Let us assume, D1 sold "x" window inverter ACs => Number of split inverter ACs sold is $13-x$

From condition-4

c) Number of split ACs sold by D1 will be "2x"

From condition-5

d) Let us assume 'y' is the number of window ACs sold by D3 & D4 => D2 sold $3y$ ACs of this type.

From condition-7

e) Let us assume 'z' is the number of split inverter ACs sold by D3 and D4 => D2 sold $2z$ ACs of this type.

Let us use a, b, c, d, and e make a table:

D1 Total =			
Split =		Window = x	
Inv = 13-x	Non-Inv =	Inv = x	Non-Inv = 0

D2 Total =			
Split =		Window = 3y	
Inv = 2z	Non-Inv =	Inv =	Non-Inv = 4

D3 Total =			
Split =		Window = y	
Inv = z	Non-Inv = 3	Inv =	Non-Inv = 2

D4 Total =			
Split =		Window = y	
Inv = z	Non-Inv =	Inv =	Non-Inv = 0

We know that the total number of window ACs is 15

$\Rightarrow x + 3y + y + y = 15 \Rightarrow x + 5y = 15$, also x and y should be greater than or equal to 2 from condition-1

$\Rightarrow x = 5$ and $y = 2$ is the only solution.

Filling this in the table:

D1 Total =			
Split =		Window = 5	
Inv = 8	Non-Inv =	Inv = 5	Non-Inv = 0

D2 Total =			
Split =		Window = 6	
Inv = 2z	Non-Inv =	Inv = 2	Non-Inv = 4

D3 Total =			
Split =		Window = 2	
Inv = z	Non-Inv = 3	Inv = 0	Non-Inv = 2

D4 Total =			
Split =		Window = 2	
Inv = z	Non-Inv =	Inv = 2	Non-Inv = 0

Now, Number of split inverter ACs is 36

$\Rightarrow 8 + 2z + z + z = 36 \Rightarrow 4z = 28 \Rightarrow z = 7$.

Filling this and using (5), the number of split AC's sold by D1 is $2 \times 5 = 10$.

D1 Total = 15			
Split = 10		Window = 5	
Inv = 8	Non-Inv = 2	Inv = 5	Non-Inv = 0

D2 Total =			
Split =		Window = 6	
Inv = 14	Non-Inv =	Inv = 2	Non-Inv = 4

D3 Total = 12			
Split = 10		Window = 2	
Inv = 7	Non-Inv = 3	Inv = 0	Non-Inv = 2

D4 Total =			
Split =		Window = 2	
Inv = 7	Non-Inv =	Inv = 2	Non-Inv = 0

If D3 and D4 sold equal number of AC's, the table will look as follows:

D1 Total = 15			
Split = 10		Window = 5	
Inv = 8	Non-Inv = 2	Inv = 5	Non-Inv = 0

D2 Total = 21			
Split = 15		Window = 6	
Inv = 14	Non-Inv = 1	Inv = 2	Non-Inv = 4

D3 Total = 12			
Split = 10		Window = 2	
Inv = 7	Non-Inv = 3	Inv = 0	Non-Inv = 2

D4 Total = 12			
Split = 10		Window = 2	
Inv = 7	Non-Inv = 3	Inv = 2	Non-Inv = 0

Number of non-inverter ACs sold is $1 + 4 = 5$

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36. **B**

$$v_1 > 0.5$$

$$v_m < 1$$

As we cannot empty a blue vessel into a white vessel unless there is enough space to hold all the volume in blue vessel in the white vessel, we have to use m white vessels. This is because we cannot empty more than 1 blue vessel into 1 white vessel.

The minimum volume that can be left in each white vessel is $1 - v_m$.

=> The minimum volume that can be left in m white vessels is $m(1 - v_m)$, which is the least upper bound.

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37. **B**

To find the limiting value, let us consider that all the blue vessels are of 1 litre capacity.

If all the blue vessels have 1 litre capacity, then the number of white vessels required is smallest integer greater than or equal to $(m/2)$.

But, blue vessels are all not of the same capacity => the above value of n_1 is the upper bound.

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